



DATerra: Boundary and structure aware domain adaptation for cross-regional terraced parcel extraction

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ABSTRACT

Irregular agricultural parcels, particularly terraced fields, exhibit complex geometries and fine-scale boundaries that are highly sensitive to cross-regional variations in imaging conditions and landscape patterns. Although unsupervised domain adaptation (UDA) has shown promise for remote sensing segmentation under domain shifts, most existing methods primarily focus on semantic alignment while failing to adequately preserve boundary geometry and parcel-level structural consistency, often leading to boundary adhesion and distorted parcel delineation in target domains. To address these limitations, we propose DATerra, a boundary and structure aware domain adaptation framework for cross-regional terraced parcel extraction. The framework couples two architecture-level boundary-modeling modules with two training-level structure-preserving strategies: (i) Boundary-Guided Attention Fusion (BGAF), which injects geometry-aware cues into multi-scale feature learning; (ii) Boundary-aware Uncertainty-guided Gating (BUDG), which selectively refines ambiguous boundary regions while preserving reliable parcel interiors; and (iii) Parcel Structural Consistency (PSC), together with Structure-Aware ClassMix (SA-ClassMix), which promotes stable parcel predictions under partial target-domain observations and preserves parcel-boundary structural units during cross-domain mixing. Extensive experiments on the GTPBD-China benchmark, covering four geographic regions and twelve cross-domain transfer scenarios, demonstrate that DATerra substantially improves boundary delineation and object-level parcel integrity. On average, DATerra improves OIS and ODS by 14.00 and 10.76 percentage points, respectively, while reducing GUC and GTC by 20.14 and 9.98 points compared with the runner-up methods. Pixel-level metrics are also consistently improved, with an average gain of 1.23%. These results highlight the importance of explicitly modeling boundary geometry and structural consistency for robust cross-regional terraced parcel extraction.

1. Introduction

Accurate agricultural parcel delineation is fundamental to crop monitoring, yield estimation, and precision management, as parcel boundaries define the basic spatial units for downstream agricultural analysis (Pan et al., 2023; Weiss et al., 2020). In mountainous regions, terraced fields are particularly important for slope farming, soil and water conservation, and sustainable agricultural development (Ferro-Vázquez et al., 2017). However, terraced parcels are typically narrow,

curved, fragmented, and strongly constrained by terrain, resulting in weak boundaries and highly irregular spatial layouts that make accurate delineation particularly challenging. Although recent studies have mainly focused on crop characterization, temporally transferable stacked ensemble learning and SAR-optical data fusion have demonstrated the value of complementary observations for mitigating temporal and sensor-induced variability (Majnoun Hosseini et al., 2024; Hamidi et al., 2024). These strategies are also relevant to cross-regional terraced mapping under heterogeneous imaging conditions.

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Very-high-resolution (VHR) remote sensing imagery and deep learning have substantially advanced agricultural parcel mapping by providing fine spatial details and automated feature extraction (Persello et al., 2019; Long et al., 2022). Existing studies have advanced field delineation through semantic segmentation, boundary-aware learning, edge-guided supervision, and hybrid representation models (Long et al., 2022; Lu et al., 2024; Zhao et al., 2025; Wang et al., 2022; Chen et al., 2024; W. Wu et al., 2025; Zhao et al., 2026). More recent work has further emphasized explicit boundary and structural modeling through terraced-field recognition, boundary-guided CNN-Transformer architectures, contour-texture hierarchical fusion, boundary enhancement, and frequency- and shape-aware feature learning (Xie et al., 2024; H. Wu et al., 2025; Xie et al., 2025a,b, 2026; Zhang et al., 2026; Huang et al., 2026). Public datasets such as FHAPD, GloCAB, AI4Boundaries, and FTW have also supported systematic evaluation of crop mapping and agricultural field boundary extraction (Zhao et al., 2025; Hall et al., 2024; d'Andrimont et al., 2023; Kerner et al., 2025). Nevertheless, these boundary-aware approaches are largely developed under supervised single-domain settings, while available datasets predominantly represent general or relatively regular agricultural landscapes, where parcels are larger, flatter, and more clearly separated. Consequently, their effectiveness in preserving fine-scale boundaries and parcel-level structural integrity under cross-regional domain shifts in terraced landscapes remains insufficiently investigated.

The recent release of GTPBD (Zhang et al., 2025) provides fine-grained annotations for terraced parcel and boundary extraction, including pixel-level masks, boundary labels, and object-level parcel labels. This dataset makes systematic research on complex terraced parcel extraction possible. Nevertheless, large-scale terraced parcel mapping still faces a severe annotation bottleneck. Earth Observation applications inevitably involve images acquired from different regions, seasons, acquisition dates, illumination conditions, and cultivation stages. It is impractical to annotate dense parcel and boundary labels for every target region, especially in terraced landscapes where manual delineation is labor-intensive and requires careful interpretation of weak and irregular boundaries. Therefore, improving target-domain performance without additional target annotations is essential for scalable terraced parcel extraction (Zheng et al., 2026).

This makes cross-regional terraced parcel extraction an important but challenging problem. Terraced regions may differ substantially in radiometric responses, texture patterns, terrain relief, slope orientation, shadow effects, parcel morphology, and cultivation practices (Z. Liu et al., 2024). Such differences cause strong domain shifts and hinder direct model transfer across regions. Unsupervised domain adaptation (UDA) provides a practical solution by transferring models from labeled source regions to unlabeled target regions (Zhu et al., 2023; Q. Li et al., 2025; H. Li et al., 2026). However, existing UDA methods mainly reduce appearance- or feature-level discrepancies through semantic alignment (Hoyer et al., 2022a,b; Chen et al., 2023; Wei et al., 2025; Chen et al., 2025). These strategies can improve pixel-level transferability but often fail to preserve the structural coupling between parcel interiors and enclosing boundaries. For terraced parcels, semantic correctness does not necessarily imply structural correctness: small boundary errors may lead to adhesion, under-segmentation, fragmentation, and distorted parcel instances (Li et al., 2023; Yan and Roy, 2014; Liang et al., 2025).

To address this challenge, we propose **DATerra**, a boundary- and structure-aware domain adaptation framework for cross-regional terraced parcel extraction. DATerra is built on a simple structural prior: coherent parcel interiors should be enclosed by reliable boundaries. Specifically, DATerra enhances geometry-sensitive representations through boundary-aware feature modeling, refines ambiguous regions with uncertainty-aware boundary modulation, and preserves parcel-boundary structures during cross-domain training. By jointly enhancing boundary geometry and parcel-level structural consistency,

DATerra improves boundary fidelity, parcel separation, and cross-regional transferability without requiring target-domain annotations.

We evaluate DATerra on GTPBD-China across twelve cross-regional transfer scenarios covering four representative terraced regions. The main contributions are summarized as follows:

1. We formulate cross-regional terraced parcel extraction as a structure-sensitive UDA problem, where robust transfer requires preserving weak boundary geometry and parcel-level structural integrity beyond semantic alignment.
2. We propose DATerra, which couples architecture-level boundary modeling with training-level structure-preserving adaptation. BGAF and BUDG enhance and selectively refine weak boundary evidence, whereas PSC and SA-ClassMix regularize target-domain predictions and preserve parcel-boundary units during cross-domain supervision transfer.
3. We conduct comprehensive evaluations across twelve GTPBD-China transfer scenarios and six unseen cross-country scenes, demonstrating improved pixel-, boundary-, and object-level performance in cross-regional adaptation and robust performance across geographically diverse unseen landscapes.

2. Study area and datasets

This study uses the GTPBD (Global Terraced Parcel and Boundary Dataset) (Zhang et al., 2025) as the data source and constructs a cross-regional subset over representative terraced landscapes in China. GTPBD provides 0.5–0.7 m imagery with three levels of annotations, including pixel-wise masks, pixel-wise boundaries, and object-level parcels. Tiles are produced at 512×512 to balance geometric detail and contextual semantics. To capture diverse terrain, climate, and farming regimes, we select four regions from GTPBD-China, including Northwest (NW), Northeast (NE), Southwest (SW), and East China (EC), as shown in Fig. 1. These regions cover different terraced landscapes, from step-like slope farming in arid and semi-arid zones to narrow-ridge terraces in humid mountains, resulting in substantial spectral, textural, and geometric variations.

For each geographic domain, the original large scenes were spatially partitioned into mutually exclusive training, validation, and test areas before tiling. Consequently, tiles belonging to different subsets do not overlap geographically. UDA optimization uses only labeled source-domain training samples and unlabeled target-domain training images. The target-domain validation set is used only for periodic evaluation and checkpoint selection and is never involved in gradient computation, while the spatially independent target-domain test set remains completely held out until final evaluation.

To visually demonstrate the difficulty of terrace extraction and highlight the unique challenges compared to regular field scenarios, we provide a compact visual comparison with representative patches from four regions in China in Fig. 2.

Each figure consists of three columns, from left to right: the VHR image, parcel label and boundary label. This comparison emphasizes the complex and irregular boundary patterns found in terraced fields, with narrow and winding ridges, as well as discontinuous edges. These characteristics are much more complicated compared to regular agricultural fields, where boundaries are typically more continuous and uniform.

To further quantify cross-domain heterogeneity beyond visual inspection, we compute descriptive statistics over all annotated parcels. Fig. 3(a) reports the parcel pixel density in each domain, characterized by the mean, median and standard deviation of parcel pixel counts. Northeast and Northwest exhibit larger and more variable parcel sizes, while East China and Southwest are dominated by smaller and more fragmented parcels, which leads to higher boundary-to-area ratios and more challenging delineation. Fig. 3(b) shows the number of tiles in the

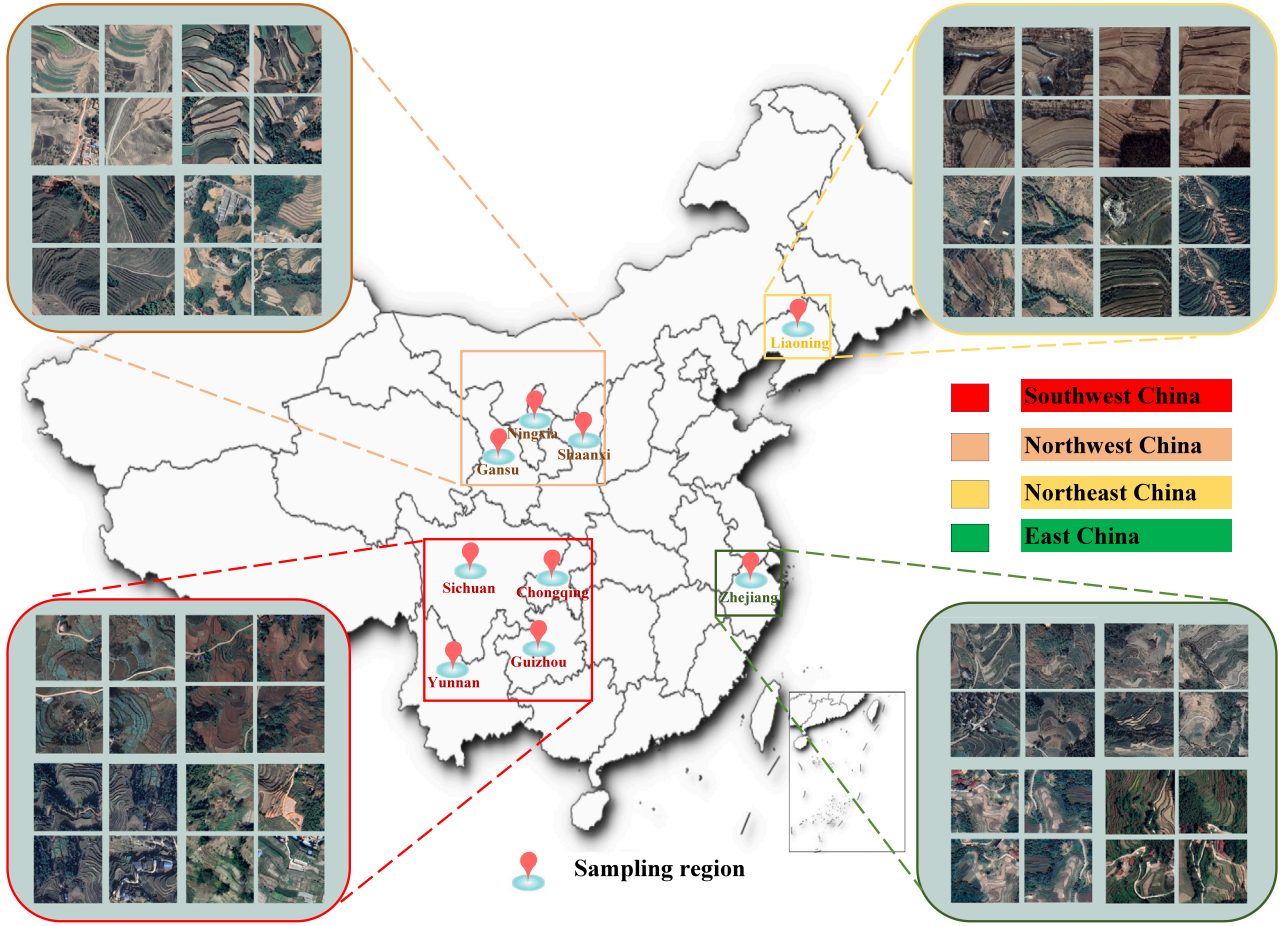


Fig. 1. Selected regions from GTPBD-China, including Northwest, Northeast, Southwest, and East China, which represent diverse terraced landscapes across China.

train, validation and test splits for each domain. The splits are intentionally kept imbalanced, which reflects the realistic situation where some terraced regions are densely sampled while others remain data scarce, and provides a natural test bed for evaluating the robustness of cross-domain adaptation from data-rich to data-poor regions.

Fig. 3(c) illustrates the per-channel mean and standard deviation of the RGB values, revealing clear differences in illumination conditions and surface reflectance patterns among different regions. These spectral discrepancies are largely driven by terrain-induced shadows, soil background and vegetation types, and they directly contribute to cross-domain appearance shifts. Fig. 3(d) presents the distribution of parcel scales in terms of pixel counts for each domain, which follows a heavy-tailed pattern and differs noticeably across regions, with some domains containing many small and narrow parcels and others dominated by larger patches. Together, these statistics highlight that our four-domain subset of GTPBD-China exhibits pronounced differences in parcel geometry, spectral characteristics and sample size, making it a suitable benchmark for studying robust UDA-based agricultural parcel extraction in complex terraced landscapes.

3. Methodology

This section details the proposed DATerra, a framework designed to tackle the geometric irregularities and domain shifts inherent in terraced parcel extraction.

3.1. Problem formulation and overall architecture

We study unsupervised domain adaptation (UDA) (Zhu et al., 2023; Ma et al., 2024) for cross-regional terraced parcel extraction. Given

a labeled source domain $\mathcal{D}_s = \{(x_s^i, y_s^i)\}_{i=1}^{N_s}$ and an unlabeled target domain $\mathcal{D}_t = \{x_t^j\}_{j=1}^{N_t}$, where x_s^i and x_t^j are remote sensing images from different regions and y_s^i is the source parcel mask, the goal is to learn a segmentation model that generalizes to the target domain without target labels. Due to differences in terrain, illumination, cultivation patterns, and imaging conditions, the two domains usually follow different data distributions, i.e., $P_s(x_s, y_s) \neq P_t(x_t, y_t)$. The ideal target-domain objective can be written as:

$$f_\theta^* = \arg \min_{f_\theta} \mathbb{E}_{(x_t, y_t) \sim P_t} [\mathcal{L}(f_\theta(x_t), y_t)]. \quad (1)$$

Like standard UDA methods, DATerra assumes labeled parcel data in the source domain, but does not require parcel or boundary annotations for the unlabeled target domain during adaptation or inference. Source-domain boundary supervision can be generated from existing parcel annotations, either by constructing boundary buffers from parcel polygons or by extracting contours from raster parcel masks. During target-domain adaptation and inference, only RGB imagery is provided.

Based on this setting, we propose DATerra, as shown in Fig. 4. The framework adopts a multi-scale encoder-decoder architecture with a Mix Transformer (MiT) encoder (Xie et al., 2021). However, standard encoder-decoder models and common UDA strategies are not sufficient for terraced parcel extraction because they mainly optimize semantic alignment, while weak boundary cues and the structural relationship between parcel interiors and enclosing boundaries may be lost under domain shifts.

To address this issue, DATerra combines boundary-aware modeling with structure-preserving adaptation. At the architectural level, Boundary-Guided Attention Fusion (BGAF) enhances geometry-sensitive features, and Boundary-Aware Uncertainty-Guided Gating (BUDG) selectively refines ambiguous boundary regions while preserving reliable

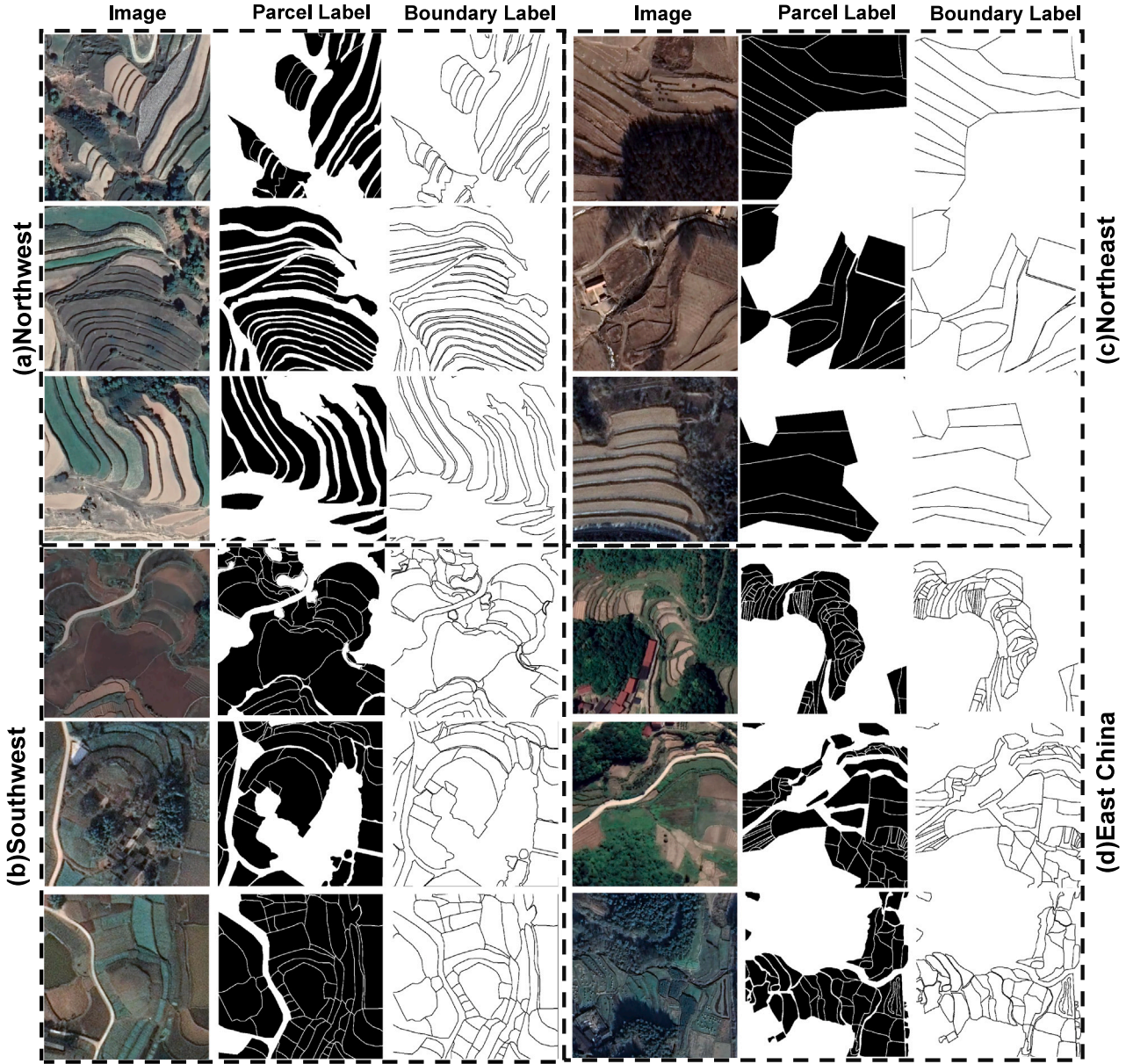


Fig. 2. Visual comparison of representative patches from four GTPBD-China regions. Each panel shows the VHR image, parcel label, and boundary label, highlighting the narrow, winding, and discontinuous boundaries of terraced parcels.

parcel interiors. At the training level, a teacher–student framework is used with Parcel Structural Consistency (PSC) and Structure-Aware ClassMix (SA-ClassMix). PSC encourages stable parcel predictions from partially masked target images, whereas SA-ClassMix transfers complete parcel–boundary structural units across domains to provide structurally coherent supervision.

3.2. Boundary-aware feature enhancement and refinement

Conventional multi-scale features are mainly optimized for semantic discrimination and may under-represent weak yet critical geometric cues along terraced parcel boundaries, especially under cross-domain appearance shifts. To address this issue, BGAF explicitly extracts boundary information from the input image as a geometric prior and uses it to guide channel- and spatial-wise attention, thereby enhancing boundary-sensitive feature representations.

Given an input image $x \in \mathbb{R}^{C \times H \times W}$, we first apply multi-directional Sobel operators (Sobel and Feldman, 1968) to capture local gradients,

followed by a lightweight convolution layer to obtain the boundary feature F_b . Based on F_b , we generate a channel attention map A_c using global pooling and a shared Multi-Layer Perceptron (MLP), and a spatial attention map A_s using channel-wise pooling and convolution:

$$\begin{aligned} F_b &= \text{Conv}_{3 \times 3}(S(x)), \\ A_c &= \sigma(\Phi(\mathcal{P}_{\text{avg}}(F_b)) + \Phi(\mathcal{P}_{\text{max}}(F_b))), \\ A_s &= \sigma(\text{Conv}_{7 \times 7}([\mathcal{P}_{\text{avg}}^c(F_b); \mathcal{P}_{\text{max}}^c(F_b)])). \end{aligned} \quad (2)$$

The attention maps are then applied to the multi-level backbone features $\{F_i\}_{i=1}^4$ to highlight boundary-related responses:

$$\tilde{F}_i = \mathcal{U}(F_i) \odot A_c \odot A_s, \quad i \in \{1, \dots, 4\}, \quad (3)$$

where $S(\cdot)$ denotes the Sobel operator, $\Phi(\cdot)$ is the shared MLP, $\mathcal{P}_{\text{avg}}$ and $\mathcal{P}_{\text{max}}$ denote average and max pooling, and the superscript c indicates channel-wise pooling. $\mathcal{U}(\cdot)$ denotes bilinear upsampling for spatial alignment, $[\cdot; \cdot]$ denotes concatenation, $\sigma(\cdot)$ is the sigmoid function, and $\odot$ is element-wise multiplication.

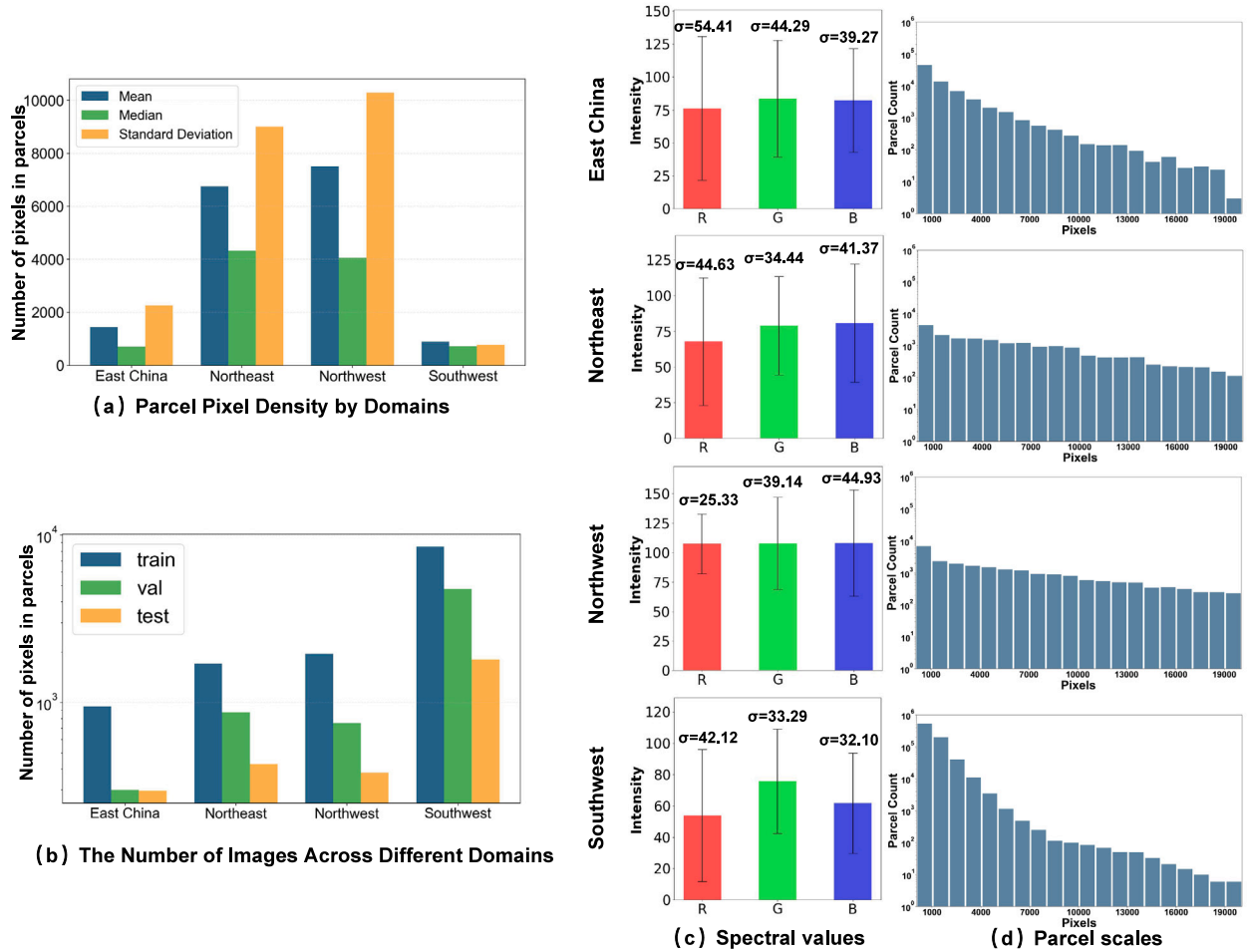


Fig. 3. Domain statistics of the four selected GTPBD-China regions. (a) Parcel pixel density. (b) Number of images in train, validation, and test splits. (c) RGB spectral statistics. (d) Parcel scale distribution. These statistics show clear geometric, spectral, and sample-size heterogeneity across different terraced regions.

Finally, we fuse the original and boundary-enhanced features through a residual connection at each hierarchical level, and concatenate the refined features as the decoder input:

$$\mathbf{F}'_i = \mathcal{U}(\mathbf{F}_i) + \gamma_i \tilde{\mathbf{F}}_i, \quad i \in \{1, \dots, 4\},$$

$$\mathbf{F}_{\text{out}} = [\mathbf{F}'_1; \mathbf{F}'_2; \mathbf{F}'_3; \mathbf{F}'_4].$$
(4)

where γ_i is a learnable scalar for balancing semantic and boundary-enhanced features.

3.2.1. Boundary-aware uncertainty-guided gating (BUDG)

Existing boundary-aware refinement strategies often apply uniform correction based on boundary strength, without distinguishing ambiguous boundary pixels from reliable parcel interiors. Such one-size-fits-all refinement may over-correct confident regions while still failing to resolve uncertain boundaries. To address this issue, we design Boundary-Aware Uncertainty-Guided Gating (BUDG), which dynamically routes predictions between parcel segmentation and boundary-aware refinement according to pixel-wise boundary cues and semantic uncertainty. Conceptually, BUDG can be viewed as a pixel-wise Mixture-of-Experts (MoE) (Shazeer et al., 2017), where the original parcel prediction branch and the boundary-aware refinement branch serve as two experts, and the gating function acts as the router. The pipeline is illustrated in Fig. 4(d).

BUDG constructs a gating map G by jointly considering the boundary probability map $B = \hat{y}_b$ and the semantic uncertainty map U . The boundary map is upsampled to the resolution of the segmentation logits $\hat{y}_{\text{seg}} \in \mathbb{R}^{C \times H \times W}$ and normalized by a sigmoid function. For

uncertainty estimation, we adopt a margin-based measure, where p_{max} and p_{second} denote the highest and second-highest class probabilities from the softmax-normalized segmentation output:

$$U = 1 - (p_{\text{max}} - p_{\text{second}}), G = \sigma(\text{Conv}_{1 \times 1}([B; U])).$$
(5)

A larger U indicates higher prediction ambiguity, while a larger G assigns more weight to boundary-aware refinement; pixels with low G mainly preserve the original parcel prediction.

The final prediction is obtained by pixel-wise weighted fusion of two expert outputs. The parcel expert directly uses the original segmentation logits, whereas the boundary expert applies a lightweight residual refinement network f_{refine} to adjust ambiguous boundary regions:

$$\begin{aligned} Z_{\text{parcel}} &= \hat{y}_{\text{seg}}, \\ Z_{\text{boundary}} &= \hat{y}_{\text{seg}} + f_{\text{refine}}([\hat{y}_{\text{seg}}; \hat{y}_b]), \\ \hat{y}_{\text{gated}} &= (1 - G) \odot Z_{\text{parcel}} + G \odot Z_{\text{boundary}}. \end{aligned}$$
(6)

Here, f_{refine} consists of two 3×3 convolutional layers with batch normalization and ReLU activation. In this way, BUDG preserves reliable parcel-interior predictions while selectively refining uncertain boundary regions.

3.3. Structure-preserving cross-regional training

3.3.1. Parcel Structural Consistency (PSC)

Inspired by Masked Image Consistency (Hoyer et al., 2023), we introduce a parcel-oriented variant termed Parcel Structural Consistency

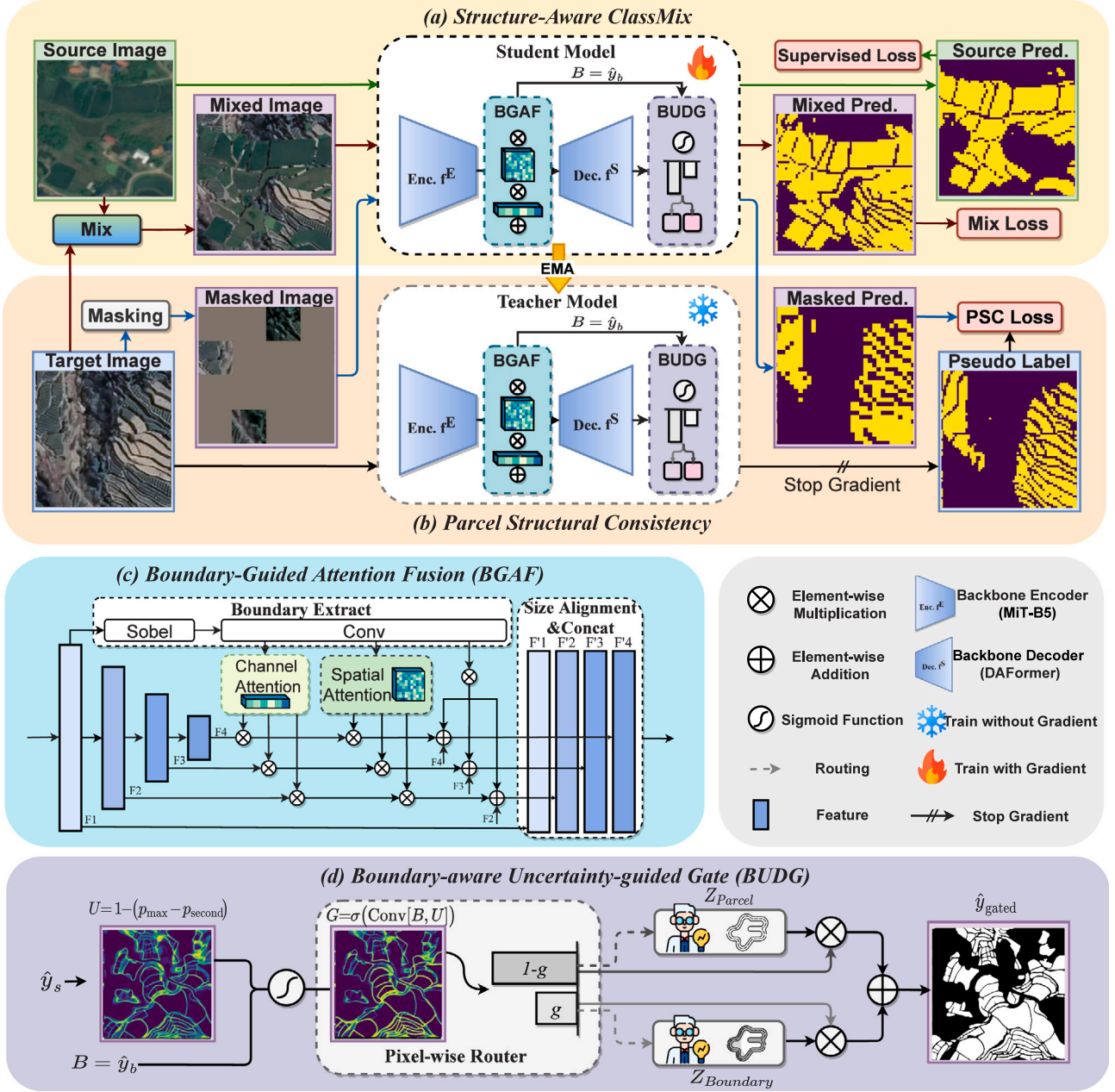


Fig. 4. Overview of DATerra. The framework consists of two boundary-aware modeling modules, i.e., BGAF and BUDG, together with two structure-preserving training strategies, i.e., PSC and SA-ClassMix, under a teacher-student UDA paradigm.

(PSC). The key intuition is that a model that has truly internalized structural knowledge should be able to infer the semantic category of occluded regions by relying solely on the surrounding contextual information, even when local visual evidence is missing. This contextual regularization helps improve the spatial coherence and structural completeness of parcel predictions under domain shifts.

For an unlabeled target image x_t , the EMA teacher g_ϕ predicts a pseudo label from the complete image:

$$\hat{y}_t = \text{Softmax}(g_\phi(x_t)).$$

Meanwhile, a high-rate random patch mask is applied to obtain $x_{\text{mask}} = \mathcal{M}_{\text{random}} \odot x_t$, which is fed into the student network:

$$\hat{y}_{\text{mask}} = \text{Softmax}(f_\theta(x_{\text{mask}})).$$

The PSC loss enforces consistency between the student prediction from the masked target image and the teacher pseudo-label on high-confidence pixels:

$$\mathcal{L}_{\text{psc}} = -\frac{1}{|\Omega|} \sum_{i \in \Omega} [\hat{y}_t^i \log \hat{y}_{\text{mask}}^i + (1 - \hat{y}_t^i) \log (1 - \hat{y}_{\text{mask}}^i)], \quad (7)$$

$$\Omega = \left\{ i \mid \max(\hat{y}_t^i, 1 - \hat{y}_t^i) \geq \tau \right\}.$$

This objective reduces the dependence on local appearance cues and encourages spatially consistent parcel predictions from partial observations under domain shifts.

3.3.2. Structure-aware ClassMix (SA-ClassMix)

Conventional ClassMix-based domain mixing treats semantic classes as independent pixel sets, which may break parcel-boundary continuity

in fine-grained land-parcel segmentation (Olsson et al., 2021; Tranheden et al., 2021). SA-ClassMix instead uses complete parcel-boundary units as the basic mixing primitives. To address this issue, we propose SA-ClassMix, which upgrades the basic unit of mixing from scattered class pixels to semantically consistent and structurally coherent parcel-boundary units. In this way, the content migrated during cross-domain mixing corresponds to complete parcel instances together with their associated boundaries, rather than arbitrary pixel regions.

Given a source image x_s and annotation y_s , we perform connected-component analysis (Soille, 1999) on parcel-interior pixels to obtain each parcel instance mask $\mathcal{M}_{\text{parcel}}^k$. Connected components smaller than 16 pixels are excluded to suppress noisy regions, while no upper parcel-size constraint is imposed. To obtain the boundary precisely associated with this instance, we apply a dilation operator (Serra, 1982) to $\mathcal{M}_{\text{parcel}}^k$ and intersect the result with the global boundary mask $\mathcal{M}_{\text{boundary}}$. In practice, the dilation radius is set to $r = 2$ pixels in all experiments, which provides a compact boundary neighborhood while avoiding excessive inclusion of adjacent structures. In this manner, we construct a structural unit U_k that contains a single parcel and its associated boundary neighborhood. The corresponding masks are generated as

$$\begin{aligned}\mathcal{M}_{\text{boundary}}^k &= \text{Dilation}(\mathcal{M}_{\text{parcel}}^k) \cap \mathcal{M}_{\text{boundary}}, \\ U_k &= \mathcal{M}_{\text{parcel}}^k \cup \mathcal{M}_{\text{boundary}}^k.\end{aligned}\quad (8)$$

Each U_k thus forms a structurally coherent and semantically consistent entity. During each iteration, SA-ClassMix uses all eligible parcel-boundary units in the current source image rather than sampling only a subset. These units are merged by logical union to form the final mixing mask $\mathcal{M}_{\text{mix}}$. Consequently, overlapping regions are included only once and require no additional priority rule. No explicit pixel-area quota is imposed; the mixed area is determined by the union of all eligible units. This mask is then used to embed the source-domain structural units into the target-domain image x_t . The mixed image x_{mix} and label y_{mix} are generated as

$$\begin{aligned}x_{\text{mix}} &= \mathcal{M}_{\text{mix}} \odot x_s + (1 - \mathcal{M}_{\text{mix}}) \odot x_t, \\ y_{\text{mix}} &= \mathcal{M}_{\text{mix}} \odot y_s + (1 - \mathcal{M}_{\text{mix}}) \odot \hat{y}_t,\end{aligned}\quad (9)$$

where $\odot$ denotes element-wise multiplication, and $\hat{y}_t$ is the pseudo label for the target-domain image produced by the teacher model. The same geometric mixing mask is applied jointly to the parcel and boundary targets so that each parcel and its associated boundary are transferred together, while their label tensors and supervision losses remain separate. The inserted source regions use ground-truth supervision, whereas low-confidence pseudo-labels in the remaining target regions are excluded using the confidence-filtering strategy described in Section 4.1. The mixed image x_{mix} is then fed into the student network f_θ to obtain the predicted probability $\hat{y}_{\text{mix}} = \text{Softmax}(f_\theta(x_{\text{mix}}))$. The training objective is defined as the binary cross-entropy loss $\mathcal{L}_{\text{mix}}$:

$$\mathcal{L}_{\text{mix}} = -[y_{\text{mix}} \log \hat{y}_{\text{mix}} + (1 - y_{\text{mix}}) \log (1 - \hat{y}_{\text{mix}})]. \quad (10)$$

SA-ClassMix transfers each source-domain parcel together with its associated boundary neighborhood as a unified structural unit, thereby preserving their internal correspondence during mixing. However, it does not formally guarantee boundary closure or continuity at the source-target interface.

3.4. Loss functions

As illustrated in Fig. 4, the student network is optimized using three components: supervised learning on source-domain images, the SA-ClassMix loss, and the PSC loss on masked target-domain samples. The supervised branch jointly learns parcel segmentation, boundary prediction, and uncertainty-guided gated refinement. For all binary prediction tasks, we adopt the standard Binary Cross-Entropy (BCE) loss:

$$\mathcal{L}_{\text{BCE}}(\hat{y}, y) = -[y \log \hat{y} + (1 - y) \log (1 - \hat{y})], \quad (11)$$

where $\hat{y}$ denotes the predicted probability and y is the ground-truth label. All supervised branches, including segmentation, boundary prediction, and gated refinement, are optimized using BCE loss. In particular, $\mathcal{L}_{\text{gate}}$ supervises the gated output $\hat{y}_{\text{gated}}$ against the source-domain parcel mask, encouraging the refinement branch to correct uncertain boundary predictions without altering reliable interior regions. Accordingly, the overall supervised objective on source-domain images is formulated as:

$$\mathcal{L}_{\text{sup}} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{bd}} + \mathcal{L}_{\text{gate}}, \quad \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \mathcal{L}_{\text{mix}} + \mathcal{L}_{\text{psc}}. \quad (12)$$

Here, $\mathcal{L}_{\text{mix}}$ promotes structure-aware cross-domain mixing, while $\mathcal{L}_{\text{psc}}$ enforces masked teacher-student consistency for target-domain parcel predictions. This unified objective encourages accurate parcel segmentation, precise boundary localization, and more structurally consistent adaptation under cross-domain shifts.

4. Experiments

4.1. Setup

All experiments are implemented in PyTorch on a single NVIDIA RTX 4090 GPU. Following HRDA (Hoyer et al., 2022b), we use an ImageNet-pretrained MiT-B5 encoder as the backbone. The model is optimized with AdamW, using a weight decay of 0.01, $\beta_1 = 0.9$, an initial learning rate of 6×10^{-5} , and a polynomial decay schedule with a power of 0.1. Training follows a fixed schedule of 40,000 iterations without early stopping, with a batch size of 1 for both source and target streams. Images are randomly resized with a scale ratio from [0.5, 2.0] and cropped into 512×512 low-resolution contextual views and 256×256 high-resolution detail views. Random horizontal flipping, color jittering, and Gaussian blur are applied for augmentation.

For PSC, a random block mask with a masking ratio of 0.7 and a block size of 128×128 pixels is applied to each 512×512 target crop. The pseudo-label confidence threshold is set to $\tau = 0.968$, the EMA decay rate is capped at $\alpha = 0.999$, and the base PSC loss weight is set to $\lambda_{\text{psc}} = 1.0$. No PSC-specific warm-up stage is used, and PSC is activated from the beginning of UDA training. Its effective contribution is further weighted by the proportion of pixels whose maximum teacher probability exceeds τ . If no pixel satisfies the threshold, the PSC loss is set to zero for that batch, while optimization continues with the remaining objectives. The batch is not discarded, and low-confidence pixels are not assigned pseudo-label supervision. The same optimization, augmentation, and PSC settings are used for all twelve cross-regional transfer scenarios without scenario-specific retuning. For SA-ClassMix, all eligible source parcel components are used in each iteration, components smaller than 16 pixels are excluded, and their associated boundary neighborhoods are extracted using a dilation radius of $r = 2$. Target-domain pseudo-labels are filtered using the same confidence threshold described above. The target-domain validation set is used only for checkpoint selection and is never involved in gradient-based optimization. Training loss is recorded throughout optimization, while target-domain validation mIoU is evaluated every 4000 iterations. The corresponding curves are shown in Fig. 5.

For UDA baselines, including PiPa (Chen et al., 2023), HRDA (Hoyer et al., 2022b), and DAFormer (Hoyer et al., 2022a), we follow their official MiT-based implementations. To ensure a controlled and fair comparison, all UDA methods are evaluated using the same MiT-B5 backbone, source-target data splits, closed-set RGB adaptation setting, input configuration, training schedule, and evaluation protocol. This unified setting minimizes differences caused by backbone capacity and experimental configurations, allowing the observed performance differences to be more directly attributed to the adaptation strategies. For generic segmentation and parcel extraction baselines, including Mask2Former (Cheng et al., 2022) and HBGNet (Zhao et al., 2025), we use their public implementations. All methods are evaluated under the same task definition and three-level evaluation protocol.

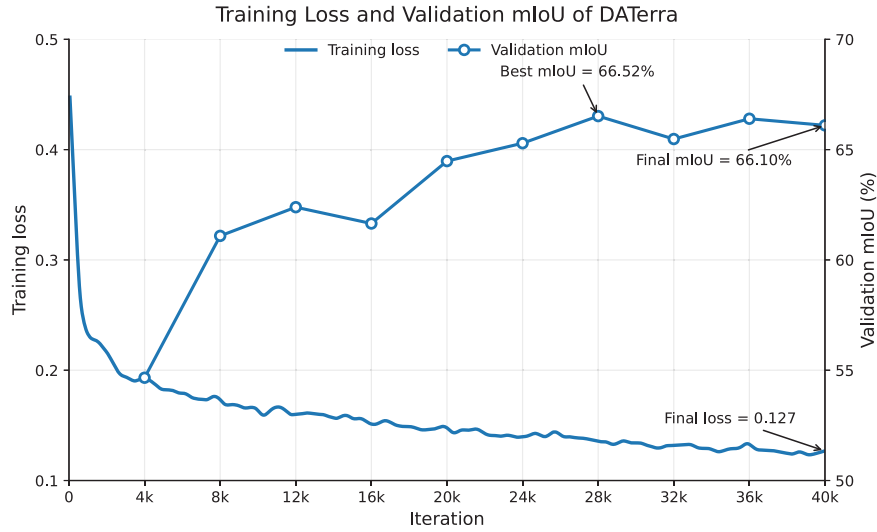


Fig. 5. Training loss and validation mIoU of DATerra under the SW→EC transfer setting over the fixed 40,000-iteration training schedule.

4.2. Three-level evaluation

Following the GTPBD-China evaluation protocol (Zhang et al., 2025), we evaluate all methods at three complementary levels: pixel level, boundary level, and object level. These metrics jointly assess parcel/background classification, boundary localization, and parcel-level geometric integrity.

Pixel-level metrics. Pixel-level performance is evaluated using F1-score, Overall Accuracy (OA), mean Intersection over Union (mIoU), and mean Accuracy (mAcc). In the binary setting, the two classes correspond to parcel and non-parcel regions. The metrics are defined as

$$\begin{aligned}
 F1 &= \frac{2TP}{2TP + FP + FN}, \\
 OA &= \frac{TP + TN}{TP + FP + FN + TN}, \\
 mIoU &= \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FP_c + FN_c}, \\
 mAcc &= \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FN_c},
 \end{aligned} \quad (13)$$

where TP , FP , FN , and TN denote true positive, false positive, false negative, and true negative pixels, respectively. C is the number of classes, and the subscript c denotes the corresponding class.

Boundary-level metrics. Boundary localization is evaluated using Optimal Dataset Scale F1-score (ODS) and Optimal Image Scale F1-score (OIS). Boundary correspondences are determined using the Euclidean-distance-transform-based matching strategy adopted by GTPBD-China. Let P_t and R_t denote boundary precision and recall at threshold t , with

$$F_t = \frac{2P_t R_t}{P_t + R_t}. \quad (14)$$

ODS selects a single optimal threshold for the entire dataset:

$$ODS = \max_{t \in \mathcal{T}} \frac{2P_t^{\text{dataset}} R_t^{\text{dataset}}}{P_t^{\text{dataset}} + R_t^{\text{dataset}}}, \quad (15)$$

whereas OIS averages the optimal F1-score of each individual image:

$$OIS = \frac{1}{N} \sum_{i=1}^N \max_{t \in \mathcal{T}} \frac{2P_t^{(i)} R_t^{(i)}}{P_t^{(i)} + R_t^{(i)}}. \quad (16)$$

Here, $\mathcal{T}$ denotes the set of evaluated thresholds, and N is the number of test images.

Object-level metrics. Object-level evaluation measures whether individual parcels are accurately separated and geometrically preserved. Let S_i denote the i th predicted parcel, and let O_i denote the ground-truth parcel with the largest overlap with S_i . The over-classification and under-classification errors are defined as

$$OC(S_i) = 1 - \frac{\text{area}(S_i \cap O_i)}{\text{area}(S_i)}, \quad UC(S_i) = 1 - \frac{\text{area}(S_i \cap O_i)}{\text{area}(O_i)}. \quad (17)$$

Their global forms are

$$GOC = \sum_{i=1}^m OC(S_i) \frac{\text{area}(S_i)}{\sum_{k=1}^m \text{area}(S_k)}, \quad (18)$$

$$GUC = \sum_{i=1}^m UC(S_i) \frac{\text{area}(S_i)}{\sum_{k=1}^m \text{area}(S_k)}, \quad (19)$$

where m is the number of predicted parcels. Global Total Classification Error (GTC) combines the two types of object-level error:

$$TC(S_i) = \sqrt{\frac{OC(S_i)^2 + UC(S_i)^2}{2}}, \quad (20)$$

$$GTC = \sum_{i=1}^m TC(S_i) \frac{\text{area}(S_i)}{\sum_{k=1}^m \text{area}(S_k)}. \quad (21)$$

The main experiments report GUC and GTC to characterize parcel under-segmentation and overall object-level geometric error.

Higher F1-score, OA, mIoU, mAcc, OIS, and ODS values indicate better performance, whereas lower GOC, GUC, and GTC values are preferred. Together, the three evaluation levels provide a comprehensive assessment of semantic accuracy, boundary fidelity, and parcel-level geometric integrity.

4.3. Comparative experiments

We evaluate DATerra on four GTPBD-China domains, including Southwest (SW), Northeast (NE), Northwest (NW), and East China (EC). Each domain is alternately used as the labeled source domain, while the remaining three domains serve as unlabeled target domains, forming 12 cross-regional transfer directions. Generic segmentation and agricultural parcel extraction baselines are evaluated under the source-only setting, whereas UDA methods use both labeled source data and unlabeled target data for adaptation.

As reported in Tables 1 and Fig. 6, DATerra achieves the best overall performance across pixel-, boundary-, and object-level metrics. Averaged over all transfers, DATerra improves OIS and ODS by 14.00% and 10.76%, respectively, and reduces GUC and GTC by 20.14% and

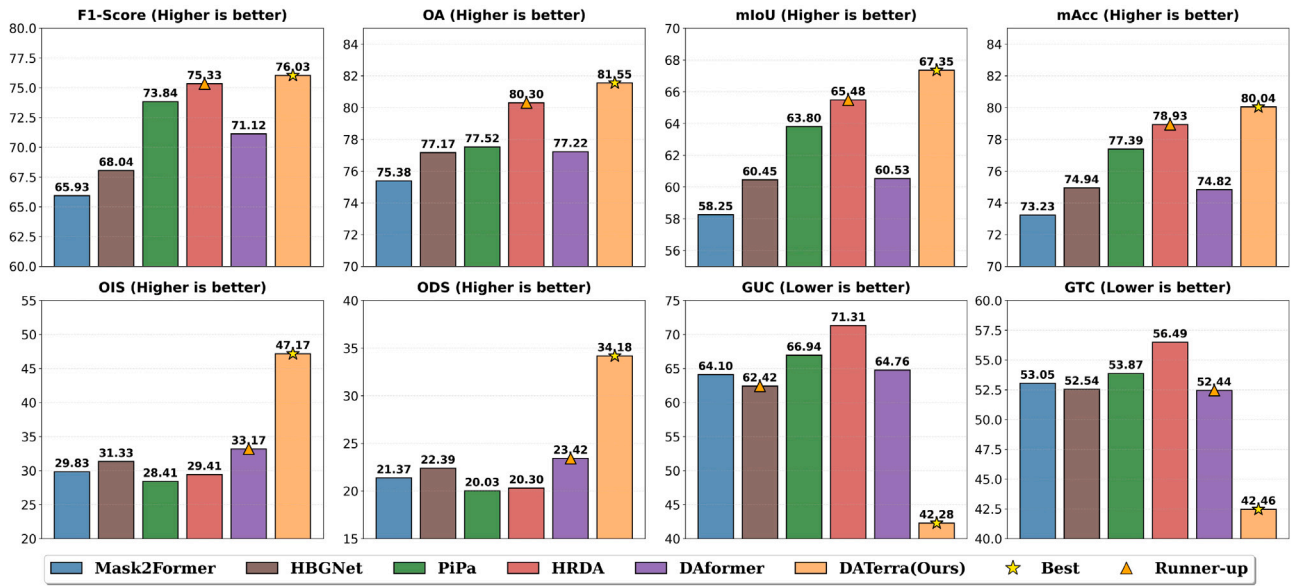


Fig. 6. Summary bar charts highlighting the overall gains of DATerra. The panel reports the mean performance across 12 transfer directions for pixel-, boundary-, and object-level metrics. Bars marked with a star denote the best-performing method, whereas triangles denote the runner-up. Higher values are better for OIS/ODS, while lower values are better for GUC/GTC.

Table 1

Cross-regional terraced parcel extraction results on twelve GTPBD-China transfer settings. Values are percentages. The best and second-best results in each setting are highlighted in bold and underline, respectively.

Domain	Model	Pixel-level				Boundary-level		Object-level	
		F1↑	OA↑	mIoU↑	mAcc↑	OIS↑	ODS↑	GUC↓	GTC↓
SW→EC	Mask2Former	46.16	66.30	45.31	62.71	38.64	27.41	51.43	52.86
	HBGNet	67.39	75.10	58.65	73.32	<u>60.34</u>	<u>46.06</u>	<u>43.89</u>	<u>50.63</u>
	PiPa	75.48	76.82	62.29	77.28	23.13	16.63	86.08	62.93
	HRDA	<u>77.26</u>	<u>77.67</u>	<u>64.74</u>	<u>78.07</u>	29.60	21.49	84.09	60.90
	DAFormer	46.95	65.68	45.10	62.33	41.31	29.31	<u>43.76</u>	51.03
	DATerra (Ours)	77.75	78.71	65.13	78.63	66.85	51.28	54.87	48.37
SW→NW	Mask2Former	46.94	75.56	51.63	64.26	36.15	25.85	60.92	57.33
	HBGNet	51.87	77.26	54.55	66.86	48.39	36.20	49.19	54.38
	PiPa	<u>62.96</u>	81.89	<u>62.27</u>	<u>73.17</u>	46.80	34.00	52.45	52.35
	HRDA	62.79	81.63	62.01	73.09	<u>48.62</u>	<u>30.56</u>	44.90	51.45
	DAFormer	52.83	77.78	55.26	67.41	46.45	32.93	46.79	54.39
	DATerra (Ours)	63.56	<u>81.72</u>	62.42	73.58	49.22	<u>35.67</u>	<u>46.73</u>	48.99
SW→NE	Mask2Former	44.05	60.15	40.49	62.58	15.52	10.41	24.24	<u>46.73</u>
	HBGNet	62.83	71.43	54.07	71.07	26.73	18.47	<u>25.18</u>	51.74
	PiPa	79.77	80.72	67.62	80.70	<u>30.32</u>	<u>21.98</u>	52.26	50.69
	HRDA	<u>81.56</u>	<u>81.66</u>	<u>69.01</u>	<u>81.68</u>	27.39	18.53	63.32	53.06
	DAFormer	68.65	68.88	52.53	69.30	28.83	19.84	59.16	54.80
	DATerra (Ours)	82.12	82.44	70.13	82.44	33.47	23.05	44.27	44.33
EC→SW	Mask2Former	82.01	81.32	68.49	82.05	<u>47.84</u>	<u>36.14</u>	<u>78.75</u>	57.39
	HBGNet	82.79	<u>82.46</u>	<u>70.15</u>	<u>83.19</u>	41.68	30.88	80.37	58.01
	PiPa	80.07	80.33	67.12	80.78	36.71	26.57	80.22	60.05
	HRDA	<u>82.95</u>	81.57	68.74	82.40	25.83	18.40	82.71	58.58
	DAFormer	80.96	82.24	69.73	82.44	46.31	34.10	80.68	<u>57.24</u>
	DATerra (Ours)	83.39	83.26	70.42	83.44	66.19	48.07	38.49	37.51
EC→NE	Mask2Former	77.22	71.01	52.94	69.39	26.31	18.10	70.77	52.93
	HBGNet	80.43	78.60	64.54	78.78	24.98	16.63	67.84	52.77
	PiPa	76.14	70.31	52.53	69.06	<u>30.79</u>	<u>21.28</u>	<u>67.01</u>	<u>52.57</u>
	HRDA	81.44	<u>79.01</u>	<u>65.94</u>	<u>79.86</u>	26.42	18.74	83.07	88.22
	DAFormer	75.57	66.47	45.53	64.10	29.60	20.78	75.27	53.31
	DATerra (Ours)	<u>81.37</u>	79.84	66.42	80.05	33.90	23.39	55.50	52.24
EC→NW	Mask2Former	73.83	81.26	66.52	83.32	<u>38.11</u>	<u>27.39</u>	78.16	58.51
	HBGNet	74.56	81.64	67.16	84.05	32.68	22.92	79.01	58.05
	PiPa	73.37	79.64	64.82	83.72	22.39	15.24	81.02	59.95
	HRDA	74.44	81.06	66.56	84.29	23.90	16.49	79.84	56.99
	DAFormer	75.96	<u>82.95</u>	68.95	<u>85.01</u>	36.10	25.56	<u>77.61</u>	<u>55.42</u>
	DATerra (Ours)	<u>75.83</u>	83.03	<u>68.68</u>	85.14	47.49	33.47	59.52	51.72

Table 1 (continued).

NE→EC	Mask2Former	58.67	72.57	53.73	69.53	22.00	16.21	70.35	53.46
	HBGNet	62.83	74.20	56.40	71.55	23.46	17.06	73.75	54.64
	PiPa	64.81	76.59	59.05	72.87	23.25	16.62	73.02	54.92
	HRDA	67.79	<u>76.67</u>	<u>60.18</u>	74.33	<u>28.21</u>	<u>20.51</u>	73.22	<u>53.01</u>
	DAFormer	66.36	76.10	59.19	73.68	26.35	18.90	72.36	54.38
	DATerra (Ours)	<u>67.78</u>	76.70	60.24	<u>74.05</u>	50.62	39.16	34.47	38.82
NE→NW	Mask2Former	63.83	78.26	59.52	70.32	<u>30.11</u>	<u>21.39</u>	78.16	58.51
	HBGNet	56.30	78.42	57.06	69.27	22.22	15.32	64.89	53.95
	PiPa	63.79	<u>81.02</u>	<u>62.02</u>	73.82	20.62	14.44	69.68	55.71
	HRDA	62.07	79.74	60.36	72.78	26.67	18.44	65.88	51.39
	DAFormer	58.16	80.09	58.95	70.40	24.41	17.16	<u>61.76</u>	<u>51.20</u>
	DATerra (Ours)	<u>63.80</u>	81.26	62.97	<u>73.62</u>	34.22	24.13	46.28	45.52
NE→SW	Mask2Former	75.96	78.60	64.46	78.22	31.32	22.40	73.23	55.04
	HBGNet	72.14	75.78	60.57	75.21	27.36	19.06	73.89	56.09
	PiPa	80.89	<u>80.57</u>	<u>67.46</u>	81.27	23.19	15.48	81.44	59.16
	HRDA	<u>80.00</u>	80.40	67.21	<u>80.59</u>	25.66	17.63	79.11	56.39
	DAFormer	74.78	78.02	63.56	77.51	23.45	16.15	75.52	57.70
	DATerra (Ours)	79.40	81.05	67.54	80.26	59.43	43.64	31.83	35.10
NW→EC	Mask2Former	43.40	67.08	45.02	62.96	25.10	18.64	46.50	43.03
	HBGNet	48.73	69.09	47.99	65.26	29.27	21.82	53.93	45.63
	PiPa	74.82	61.96	66.79	76.84	10.63	7.56	63.43	52.71
	HRDA	73.28	81.36	<u>67.23</u>	<u>79.28</u>	31.88	23.23	73.76	52.91
	DAFormer	67.88	73.10	53.87	69.77	<u>36.26</u>	<u>25.60</u>	41.36	<u>42.54</u>
	DATerra (Ours)	<u>73.62</u>	<u>80.69</u>	67.61	79.71	<u>49.32</u>	<u>37.04</u>	<u>41.62</u>	38.97
NW→NE	Mask2Former	83.40	84.80	<u>73.47</u>	<u>84.69</u>	24.32	16.11	50.81	45.58
	HBGNet	82.32	81.54	68.79	81.65	28.02	18.91	52.09	42.06
	PiPa	80.07	79.75	66.32	80.06	<u>32.00</u>	<u>21.94</u>	48.39	<u>41.72</u>
	HRDA	80.60	77.39	62.43	76.59	24.47	16.43	60.23	43.71
	DAFormer	<u>83.58</u>	83.13	71.11	83.37	27.29	18.39	51.07	42.76
	DATerra (Ours)	84.11	84.89	73.71	84.84	35.98	24.74	38.25	41.27
NW→SW	Mask2Former	75.96	78.60	64.46	78.22	31.32	22.40	73.23	55.04
	HBGNet	73.62	78.41	63.68	77.44	<u>39.85</u>	<u>29.04</u>	<u>66.50</u>	<u>50.62</u>
	PiPa	75.57	79.95	65.84	78.97	35.81	25.22	67.42	52.70
	HRDA	81.68	<u>82.76</u>	<u>70.59</u>	83.29	34.49	24.36	78.33	55.72
	DAFormer	77.57	80.65	67.14	80.08	39.83	28.25	70.80	53.14
	DATerra (Ours)	<u>81.61</u>	82.79	70.73	<u>83.28</u>	59.01	43.61	28.10	32.60

9.98%. Pixel-level metrics also show a small but consistent average gain of 1.23 %, indicating that the structural improvements do not sacrifice parcel/background classification accuracy.

The results further show clear transfer-direction sensitivity across domains. HBGNet (Zhao et al., 2025) can produce sharp boundaries in some source-rich settings, but it is prone to over-segmenting terrace-like high-frequency textures when the source data are limited. UDA



Fig. 7. Qualitative comparison on the GTPBD-China dataset with SW participating as both source and target. Columns: Image, Ground Truth, DATerra (ours), Mask2Former, HBGNet, PiPa, HRDA, DAFormer. Rows (top to bottom): SW→EC, SW→NW, SW→NE, EC→SW, NE→SW, NW→SW. DATerra preserves parcel boundaries more faithfully and mitigates over-/under-segmentation and boundary adhesion artifacts. White: parcels; black: background.

Table 2

Quantitative performance of DATerra on challenging target-domain scenes under the EC→SW transfer setting.

Scenario	F1	OA	mAcc	OIS	GUC
Case 1	79.24	82.76	77.96	71.55	21.32
Case 2	80.27	81.98	79.51	74.40	18.14

baselines, including PiPa (Chen et al., 2023), HRDA (Hoyer et al., 2022b), and DAFormer (Hoyer et al., 2022a), improve semantic transferability but still suffer from unstable boundary localization and high object-level errors under strong regional shifts. In contrast, the qualitative comparisons in Figs. 7–9 show that DATerra produces clearer boundaries, fewer adhesion artifacts, and more complete parcel masks, especially in narrow and densely distributed terraced areas. These results demonstrate the robustness of DATerra to sample imbalance and heterogeneous cross-regional domain shifts.

To further evaluate DATerra under challenging landscape conditions, we select two target-domain scenes from the GTPBD Sichuan hilly region under the EC→SW transfer setting. As shown in Table 2 and Fig. 10, Case 1 contains fragmented parcels interwoven with settlements and mixed land use, whereas Case 2 exhibits irregular parcel boundaries and weak terrace cues. DATerra maintains stable performance in both cases, achieving OA values of 82.76 and 81.98, mAcc values of 77.96 and 79.51, and OIS values of 71.55 and 74.40, respectively. The qualitative results further show that DATerra recovers

the principal terrace distribution and substantial parcel structures, providing additional evidence of its robustness under heterogeneous target-domain conditions.

4.4. Ablation experiment

As illustrated in Fig. 11, the progressive addition of the proposed components, from HRDA in column (g) to the complete DATerra in column (c), gradually improves parcel coherence and boundary quality. PSC provides modest improvements in parcel prediction, while SA-ClassMix further reduces fragmentation and adhesion by preserving parcel-boundary units during cross-domain mixing. BGAF markedly recovers thin and weak boundaries, as highlighted by the red circles, but its strong geometric responses may also introduce false or spatially expanded boundaries in texture-rich or shadowed regions. BUDG suppresses these uncertain boundary responses and preserves reliable parcel interiors, producing sharper and more complete parcel predictions.

The quantitative results in Table 3 are consistent with these visual observations. After adding BGAF, OIS/ODS increase to 66.61/50.96 on SW→EC and 65.26/47.30 on EC→SW, together with substantial reductions in GUC and GTC. Although several pixel-level metrics decrease slightly, BUDG restores pixel-level accuracy while further improving the boundary- and object-level results. The remove-one variants also show consistent degradation after removing either PSC or SA-ClassMix. Overall, DATerra achieves the best result in 14 of the 16

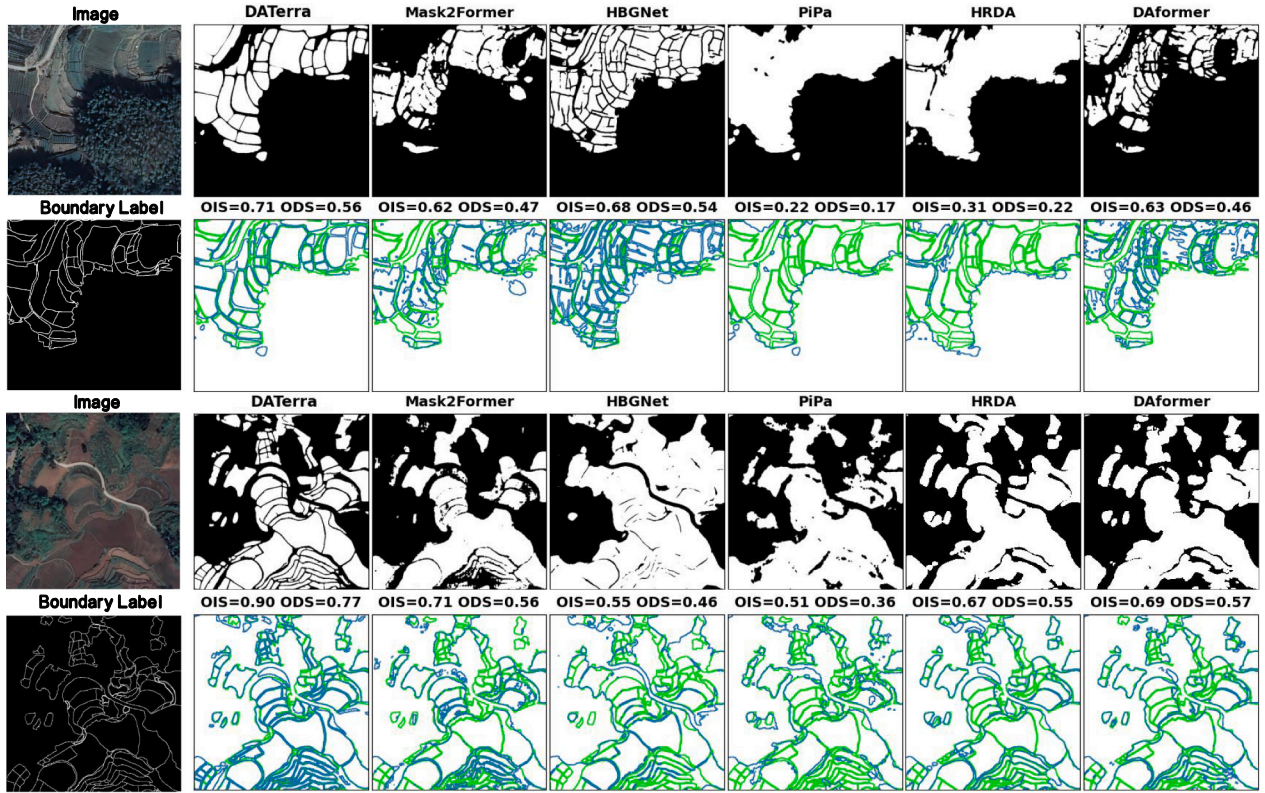


Fig. 8. Qualitative comparison of boundary-level results in the SW→EC and EC→SW transfer settings. The upper and lower groups correspond to SW→EC and EC→SW, respectively. For each case, we show the input image, the ground-truth boundary label, the predicted boundary maps of DATerra, Mask2Former, HBNet, PiPa, HRDA, and DAformer, and the corresponding boundary overlays. In the overlay visualizations, green denotes the ground-truth boundaries and blue denotes the predicted boundaries. The OIS and ODS values of each method are also reported to quantify boundary localization quality.

Table 3

Controlled, progressive, and remove-one ablations on SW→EC and EC→SW. Values are percentages. MIC denotes the standard masked-image consistency baseline. Pixel-level: F1, OA, mIoU, and mAcc; boundary-level: OIS and ODS; object-level: GUC and GTC. The best and second-best results are highlighted in **bold** and underline, respectively.

Domain	Model	Pixel-level				Boundary-level		Object-level	
		F1↑	OA↑	mIoU↑	mAcc↑	OIS↑	ODS↑	GUC↓	GTC↓
SW→EC	HRDA	77.26	77.67	<u>64.74</u>	78.07	29.60	21.49	84.09	60.90
	HRDA + MIC	77.82	78.03	64.57	78.24	32.47	24.53	77.88	57.23
	HRDA + BGAF	75.34	74.82	63.21	76.99	61.67	48.48	56.76	51.13
	HRDA + PSC	77.36	78.36	64.31	78.54	35.64	25.52	70.59	54.07
	HRDA + PSC + SA-ClassMix	77.43	78.38	64.34	<u>78.57</u>	42.27	31.12	60.71	52.90
	HRDA + PSC + SA-ClassMix + BGAF	76.00	78.28	64.39	78.27	<u>66.61</u>	<u>50.96</u>	<u>55.11</u>	<u>50.97</u>
	DATerra w/o PSC	77.39	78.32	64.23	78.44	60.31	42.56	56.61	51.28
	DATerra w/o SA-ClassMix	77.33	78.48	64.24	78.51	59.67	44.38	58.47	53.67
	DATerra(HRDA + PSC + SA-ClassMix + BGAF +BUDG)	<u>77.75</u>	78.71	65.13	78.63	66.85	51.28	54.87	48.37
EC→SW	HRDA	82.95	81.57	68.74	82.40	25.83	18.40	82.71	58.58
	HRDA + MIC	82.20	81.95	69.36	82.65	34.51	24.98	81.83	58.44
	HRDA + BGAF	81.72	80.33	66.42	80.64	55.37	40.69	45.63	42.18
	HRDA + PSC	83.04	82.93	70.34	83.21	35.37	25.63	79.67	58.30
	HRDA + PSC + SA-ClassMix	<u>83.11</u>	83.06	70.81	<u>83.43</u>	45.46	33.30	76.33	56.14
	HRDA + PSC + SA-ClassMix + BGAF	82.72	82.37	69.30	82.12	<u>65.26</u>	<u>47.30</u>	<u>39.34</u>	<u>37.54</u>
	DATerra w/o PSC	83.01	82.89	69.29	82.97	63.78	47.11	41.78	39.66
	DATerra w/o SA-ClassMix	82.97	<u>83.11</u>	69.41	82.67	61.12	44.27	42.73	40.37
	DATerra(HRDA + PSC + SA-ClassMix + BGAF +BUDG)	83.39	83.26	<u>70.42</u>	83.44	66.19	48.07	38.49	37.51

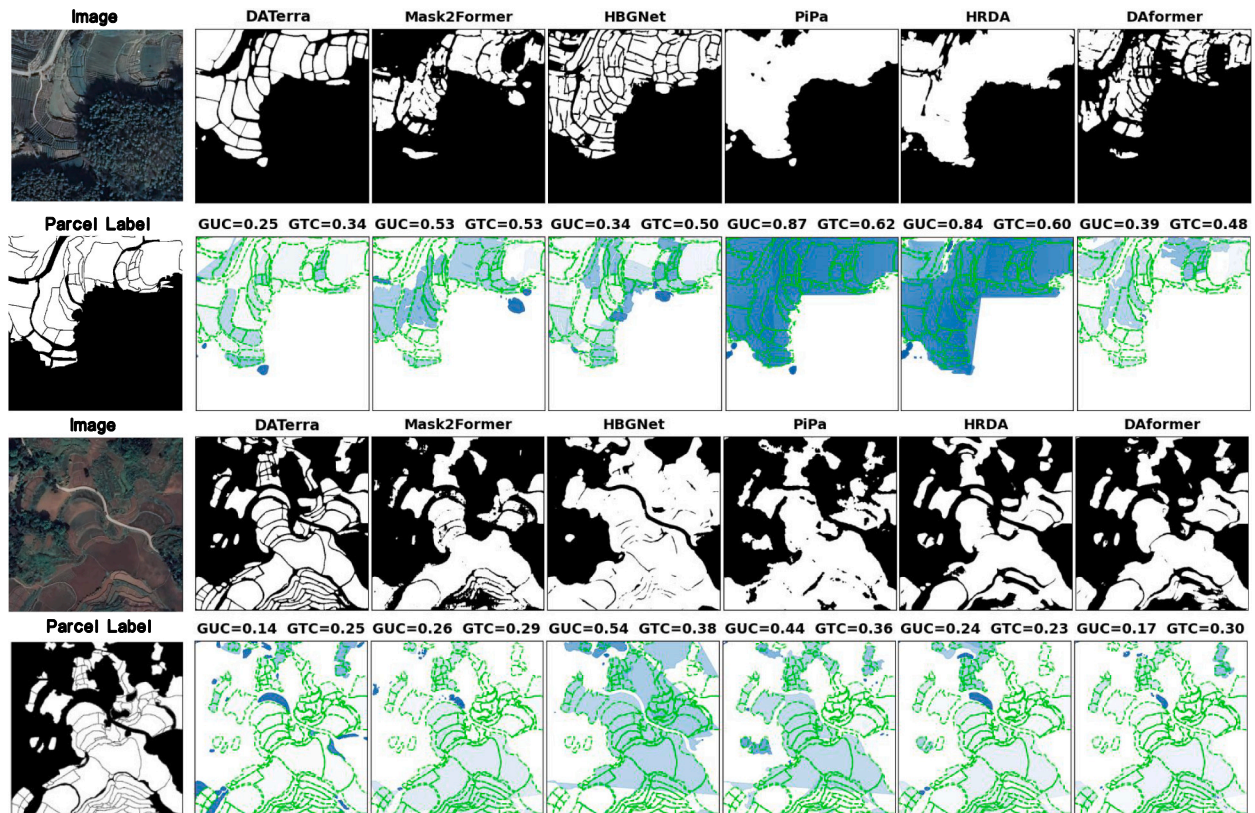


Fig. 9. Qualitative comparison of object-level results in the SW→EC and EC→SW transfer settings. The upper and lower groups correspond to SW→EC and EC→SW, respectively. For each case, we show the input image, the ground-truth parcel label, the predicted parcel maps of DATerra, Mask2Former, HBGNet, PiPa, HRDA, and DAformer, and the corresponding object-level error visualizations. In the error maps, green contours denote the ground-truth parcel boundaries, while blue indicates under-segmentation errors, with darker intensity corresponding to more severe errors. The GUC and GTC values of each method are reported to quantify object-level delineation quality.

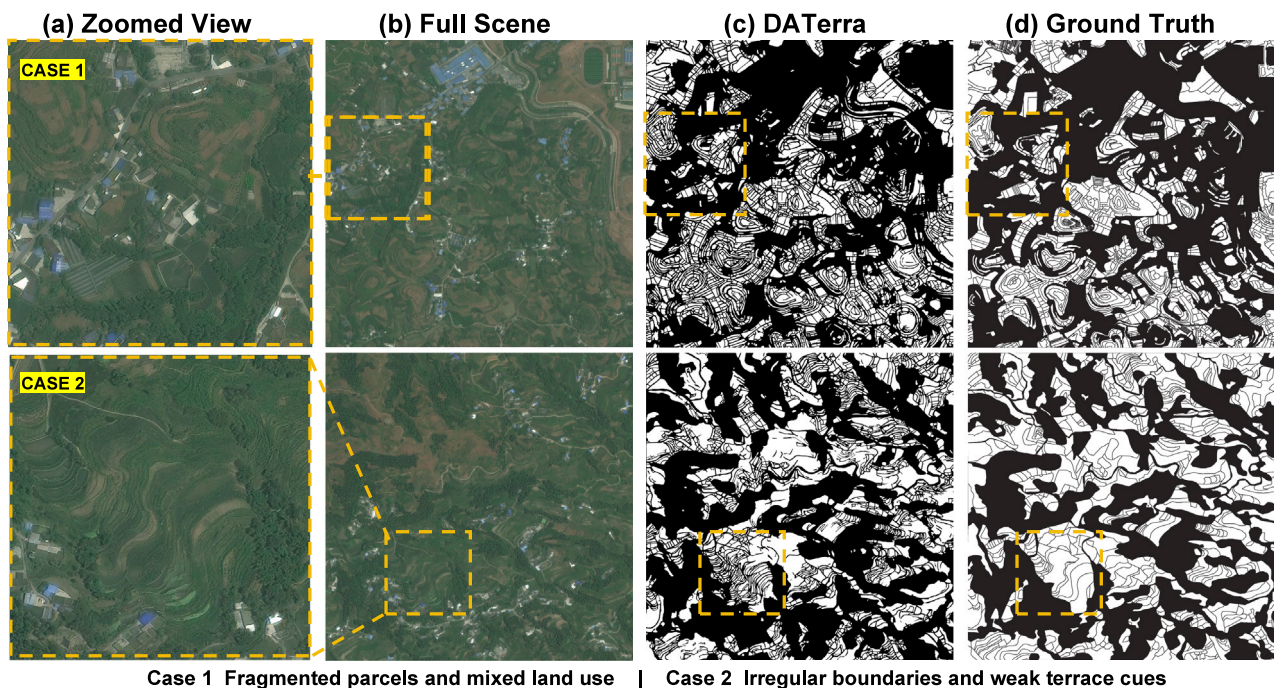


Fig. 10. Qualitative visualization of DATerra's generalization across diverse and challenging agricultural scenarios. The figure contrasts two representative landscapes: **(Case 1, Top row)** Fragmented smallholder parcels interwoven with mixed land use (e.g., residential areas and infrastructure); **(Case 2, Bottom row)** Highly irregular terraced fields characterized by weak boundary cues.

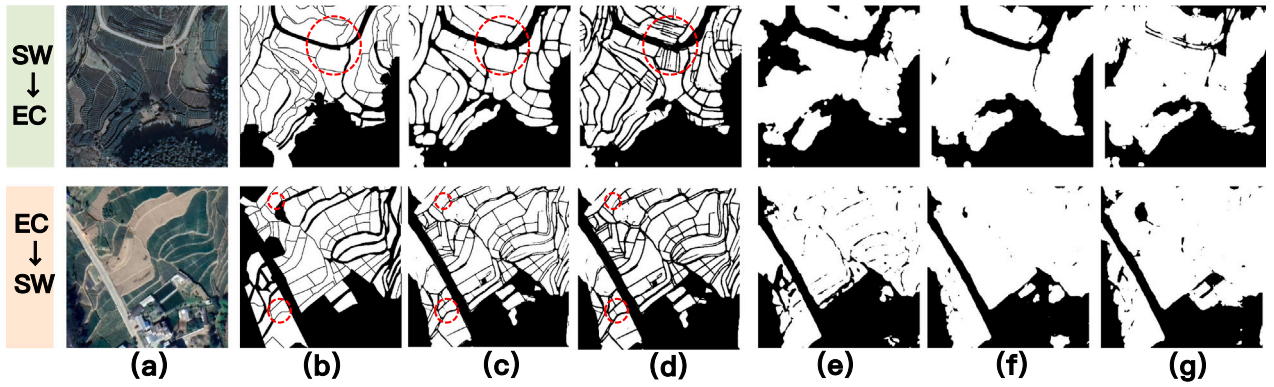


Fig. 11. Qualitative comparison across ablations: DATerra preserves sharp boundaries and parcel integrity under SW→EC and EC→SW. Columns: (a) Image, (b) Ground Truth, (c) HRDA+PSC+SA-ClassMix+BGAf+BUDG (DATerra), (d) HRDA+PSC+SA-ClassMix+BGAf, (e) HRDA+PSC+SA-ClassMix, (f) HRDA+PSC, (g) HRDA. White: parcels; black: background.

Table 4

External-benchmark UDA results on FHAPD under two cross-regional transfer directions.

Setting		F1	OA	mIoU	mAcc	OIS	GUC↓
HN→JS	HRDA	84.72	81.52	73.81	81.11	68.69	39.96
	DATerra	88.24	83.39	77.76	85.34	88.43	9.34
JS→HN	HRDA	78.55	76.40	68.78	75.57	55.25	41.88
	DATerra	84.64	82.23	78.31	83.64	85.46	8.31

domain-metric combinations and the second-best result in the remaining two, confirming that the proposed components are effective and complementary.

4.5. External-benchmark validation on FHAPD

To evaluate whether DATerra generalizes beyond the imagery and regional distributions of GTPBD-China, we conduct an external-benchmark UDA evaluation on FHAPD. Henan (HN) and Jiangsu (JS) are treated as two geographic domains, forming the HN→JS and JS→HN transfer settings. For each direction, the source-domain annotations and unlabeled target-domain images are used under the same UDA protocol, and HRDA and DATerra are evaluated using identical training and evaluation settings.

As shown in Table 4, DATerra consistently outperforms HRDA in both transfer directions. These results demonstrate that the proposed framework is not restricted to GTPBD-China and can transfer effectively across regions on an external parcel benchmark.

4.6. Computational complexity and accuracy-efficiency trade-off

We evaluate the training and inference costs of the progressive DATerra configurations under the EC→SW setting using a single NVIDIA RTX 4090, 512 × 512 inputs, and batch size 1. Training costs include the complete teacher-student UDA pipeline, while inference costs are measured on the deployed student network. As shown in Table 5, PSC and SA-ClassMix mainly increase training cost, whereas BGAf and BUDG introduce the inference overhead. Compared with HRDA, DATerra adds only 0.03M parameters and 1.3G FLOPs, with FPS decreasing from 8.80 to 8.01; training memory increases by 1.77 GB and the estimated 40,000-iteration training time by 3.00 h. In return, DATerra improves mAcc and OIS by 1.04 and 40.36 points, respectively, while reducing GUC by 44.22 points, demonstrating a favorable accuracy-efficiency trade-off.

5. Further analysis and discussion

This section positions DATerra within existing Earth Observation literature and analyzes the model behavior of BGAf, BUDG, PSC, and

SA-ClassMix. We further discuss its extensibility and limitations for large-area terraced parcel mapping.

5.1. Relation to existing earth observation studies

Agricultural parcel mapping has been advanced by semantic segmentation, boundary-aware learning, edge-guided supervision, and hybrid representation models (Zhu et al., 2024; Long et al., 2022; Li et al., 2023; Lu et al., 2024; Zhao et al., 2025; Pan et al., 2024). Public datasets such as GloCAB, AI4SmallFarms, AI4Boundaries, FTW, and GTPBD have further supported systematic evaluation of crop-field and parcel-boundary extraction (Hall et al., 2024; Persello et al., 2023; d'Andrimont et al., 2023; Kerner et al., 2025; Zhang et al., 2025). However, most existing studies focus on relatively regular arable fields, whereas terraced parcels are narrow, curved, fragmented, and strongly constrained by terrain. This makes boundary localization and object-level integrity more difficult to preserve across regions.

Beyond RGB imagery, additional spectral, temporal, and topographic observations provide complementary information for agricultural Earth Observation (Cai et al., 2023). Sentinel-2 imagery and topographic features have been used for large-scale global terrace mapping (Y. Li et al., 2025). For hyperspectral imagery, SpectralFormer (Hong et al., 2022) models local spectral sequences through group-wise spectral embeddings and cross-layer feature fusion, while SimPoolFormer (Roy et al., 2025) employs complementary spectral-spatial streams for hyperspectral image classification. These studies demonstrate the value of richer spectral and multi-source representations. However, they primarily address large-scale terrace mapping or hyperspectral image classification, whereas DATerra focuses on unsupervised cross-regional adaptation for sub-meter terraced parcel delineation, where weak-boundary localization and parcel-level structural integrity are the primary objectives.

Recent Earth Observation foundation models and large-scale pre-trained models have further improved transferability through self-supervised, multiscale, and multimodal representation learning (F. Liu et al., 2024). SatMAE models temporal and multispectral satellite observations, Scale-MAE introduces scale-aware geospatial pretraining, and CROMA learns transferable optical-SAR representations through contrastive learning and masked reconstruction (Cong et al., 2022; Reed et al., 2023; Fuller et al., 2023). DOFA further explores wavelength-conditioned representations across heterogeneous Earth Observation sensors (Liu et al., 2025), while SAM-based studies demonstrate the potential of promptable foundation models for remote sensing segmentation (Osco et al., 2023). These approaches primarily provide general pretrained representations or promptable segmentation capabilities, whereas DATerra focuses on task-specific adaptation to unlabeled target regions and explicitly addresses weak boundaries,

Table 5

Training and inference costs of DATerra under the EC→SW transfer setting. Training memory and time include the complete UDA pipeline, whereas parameters, FLOPs, and FPS are measured using the deployed student network. The total training time is estimated from the average time per iteration over 40,000 iterations. All measurements are obtained on a single NVIDIA RTX 4090 using 512×512 inputs and batch size 1. The best and second-best accuracy results are highlighted in bold and underline, respectively.

Configuration	Training cost			Inference cost			Accuracy		
	Memory (GB) ↓	Time/Iter. (s) ↓	Est. Time/40k (h) ↓	Params (M) ↓	FLOPs (G) ↓	FPS ↑	mAcc ↑	OIS ↑	GUC ↓
HRDA	8.63	0.78	8.67	85.68	247.6	8.80	82.40	25.83	82.71
HRDA + PSC	9.18	0.88	9.78	85.68	247.6	8.79	83.21	35.37	79.67
HRDA + PSC + SA-ClassMix	9.61	0.94	10.44	85.68	247.6	8.76	<u>83.43</u>	45.46	76.33
HRDA + PSC + SA-ClassMix + BGAF	10.05	0.99	11.00	85.71	247.9	8.53	82.12	<u>65.26</u>	39.34
DATerra	10.40	1.05	11.67	85.71	248.9	8.01	83.44	66.19	38.49

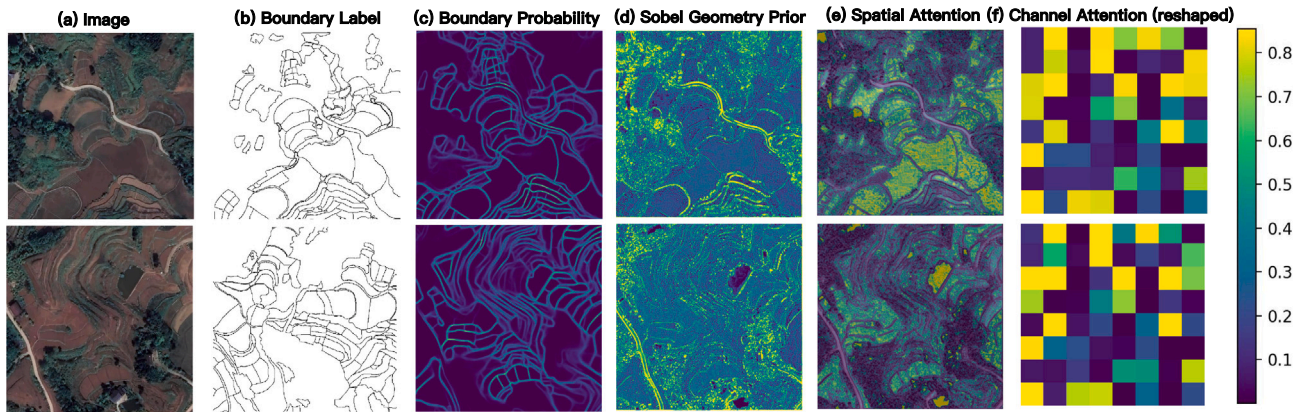


Fig. 12. Visualization of geometry-aware feature enhancement induced by BGAF. These results demonstrate that BGAF injects relatively robust candidate geometric cues and re-calibrates feature representations to recover high-frequency boundary details suppressed by domain-invariant semantic learning.

parcel adhesion, and structural fragmentation. The two directions are complementary: foundation models could serve as stronger encoders or initialization backbones for DATerra, while its boundary- and structure-aware mechanisms could continue to support target-domain adaptation. A controlled comparison and integration under the same UDA protocol remain future work.

Recent UDA methods improve remote sensing segmentation by reducing image-, feature-, or context-level domain discrepancies (Li et al., 2022; Hoyer et al., 2022a,b; Chen et al., 2023; Wei et al., 2025), and several agricultural parcel studies have explored cross-regional or cross-sensor transfer (Liu et al., 2022; Tian et al., 2024; Zhang et al., 2022). Nevertheless, these methods mainly enhance semantic transferability and rarely model the coupling between parcel interiors and enclosing boundaries. DATerra addresses this gap by formulating cross-regional terraced parcel extraction as a structure-sensitive UDA problem, explicitly preserving boundary geometry and parcel-level structural consistency for more reliable large-area Earth Observation mapping.

5.2. Recovering fine-scale terrace boundaries under domain shifts

Tables 1 shows that DATerra consistently improves boundary-related metrics over baseline UDA methods. To better understand this improvement, we visualize encoder-level feature responses in Fig. 12.

Terraced parcels contain thin, elongated, and irregular boundaries whose high-frequency geometric details are easily weakened during cross-domain adaptation. As shown in Fig. 12(c), boundary responses become blurred and fragmented compared with the ground truth in Fig. 12(b), indicating that semantic alignment may suppress weak but important geometric cues. In contrast, the Sobel-derived geometry prior in Fig. 12(d) captures relatively stable ridge-like structures despite regional appearance variations. BGAF uses these gradient-based priors to re-calibrate multi-level feature representations. As shown in Fig. 12(e–f), the spatial and channel attention responses are concentrated around narrow boundary regions and geometry-sensitive channels. This

suggests that BGAF selectively enhances geometry-sensitive cues rather than simply amplifying all local gradients. Such feature enhancement helps recover fine-scale terrace boundaries that are otherwise weakened under cross-regional feature learning.

5.3. Uncertainty-aware refinement for ambiguous parcel boundaries

Conventional post-processing methods, such as morphological operations or rigid boundary overlays, usually apply global heuristic rules and may distort correctly segmented regions (Tang et al., 2021; Zhu et al., 2022). In contrast, BUDG performs localized refinement through uncertainty-aware routing (J. Li et al., 2026). As shown in Fig. 13, uncertainty remains low in confident parcel interiors but increases around ambiguous boundary regions, where adhesion, breakage, and fragmentation are more likely to occur. BUDG generates the gating map from the concatenated boundary probability and semantic uncertainty, $[B; U]$, through a learnable 1×1 convolution and a Sigmoid function. Concatenation preserves the two cues as separate channels and allows their relative contributions to be adaptively learned. The resulting gating map assigns low values to most reliable parcel regions and activates the refinement branch primarily around ambiguous boundaries, as further illustrated by the gating-value distribution.

These results indicate that BUDG can correct local structural defects while preserving reliable parcel interiors. This behavior is important for terraced parcel mapping because over-correction may damage already well-delineated parcels, whereas under-correction may leave broken or merged terrace boundaries. Therefore, BUDG provides a more adaptive alternative to heuristic post-processing and improves boundary integrity under cross-regional domain shifts.

The ablation results in Table 3 further support this behavior. After introducing BUDG, mIoU increases from 64.39 to 65.13 on SW→EC and from 69.30 to 70.42 on EC→SW, while OIS/ODS are retained or further improved. These consistent trends indicate that uncertainty-guided routing remains effective across different regional domain shifts.

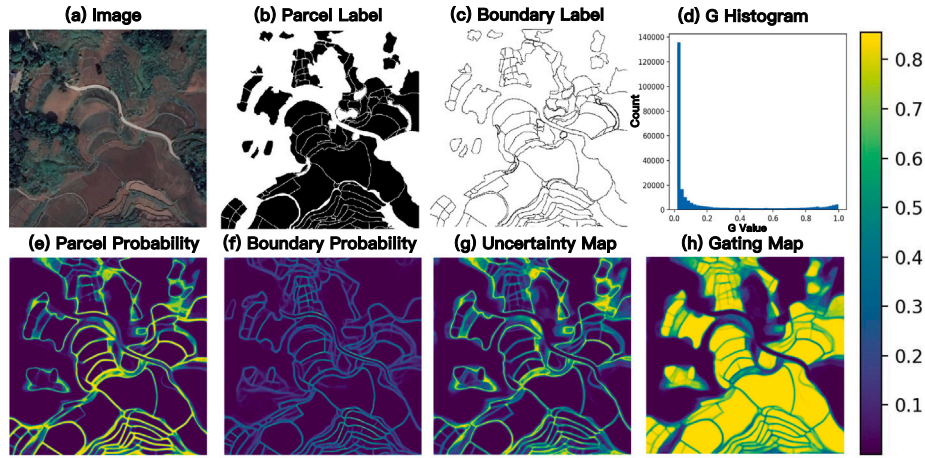


Fig. 13. Visualization of the Boundary-Aware Uncertainty-Guided Gating (BUDG) module. The figure presents (a) the input image, (b) the parcel label, (c) the boundary label, (d) the distribution of gating values, (e) the parcel probability, (f) the boundary probability, (g) the semantic uncertainty map, and (h) the learned gating map. The gating map is generated from the concatenated representation $[B; U]$ through a learnable 1×1 convolution and a Sigmoid function. Most reliable parcel regions receive low gating values, while higher values selectively activate boundary refinement in ambiguous regions.

5.4. Uncertainty sources and model reliability

Uncertainty arises at multiple stages of cross-regional terraced parcel extraction. Input imagery may be affected by illumination variation, shadows, haze, seasonal appearance changes, image blur, and spatial-resolution differences, which can weaken true parcel boundaries or introduce misleading gradients. Narrow, partially occluded, or visually ambiguous parcel boundaries may additionally contain interpretation and rasterization uncertainty. During adaptation, domain shift can lead the EMA teacher to produce erroneous or miscalibrated high-confidence pseudo-labels, and these errors may propagate through PSC and SA-ClassMix.

Regional sample-size imbalance may further induce direction-dependent pseudo-label behavior. Under the fixed 40,000-iteration schedule with a batch size of 1 for both source and target streams, regional sample counts do not directly alter the per-iteration weighting of the two domains; rather, they affect sample diversity and the frequency with which individual samples are revisited. A larger source domain may provide broader supervised coverage, whereas a smaller target domain increases the reuse of individual target samples and may repeatedly reinforce systematic high-confidence errors. Conversely, a smaller source domain may provide less comprehensive initialization, while a larger and more heterogeneous target domain may produce more variable and low-confidence predictions during early adaptation, but also offers a broader pool of reliable samples after confidence filtering. Pseudo-label quality is therefore jointly governed by sample coverage, spectral and geometric domain gaps, and target-scene complexity, rather than by sample size alone. EMA teacher smoothing, confidence filtering, and confidence-aware loss weighting mitigate these effects, although the current evaluation does not independently quantify pseudo-label noise for each transfer direction.

The Sobel-derived prior used by BGAF may also respond to shadows, roads, buildings, or texture-rich backgrounds rather than actual parcel boundaries. DATerra therefore does not treat Sobel edges as a standalone or uniformly reliable boundary detector. Instead, the raw gradient responses are transformed by a learnable convolution, selectively reweighted through channel and spatial attention, and fused with the original semantic features through learnable residual weights. BUDG further suppresses unreliable boundary responses by activating refinement primarily where boundary evidence coincides with semantic ambiguity. Thus, Sobel provides a lightweight candidate geometric cue, while the learned attention and uncertainty-aware routing determine how strongly this cue contributes to the final prediction.

5.5. From local texture matching to parcel-level structural reasoning

Tables 1 shows that DATerra consistently improves object-level metrics, indicating better preservation of parcel-level structural integrity. PSC reduces the model's reliance on local texture cues by requiring consistent predictions from partially masked target images. As shown in Fig. 14, the student network can recover spatially coherent parcel regions from incomplete observations. This behavior suggests that PSC strengthens contextual reasoning and reduces local fragmentation under domain shifts. However, PSC is a masked consistency objective and does not explicitly optimize connectivity, boundary closure, or other mathematical topological properties.

By enforcing consistency under incomplete observations, PSC strengthens long-range contextual modeling and improves parcel structural completeness. This helps reduce parcel fragmentation and boundary discontinuities under cross-regional domain shifts, thereby enhancing the structural reliability of large-area terraced parcel extraction.

5.6. Preserving parcel-boundary units during cross-domain augmentation

Effective domain adaptation for terraced parcel extraction requires preserving the structural relationship between parcel interiors and boundaries during cross-domain supervision transfer. However, conventional pixel- or region-level mixing strategies may break this relationship and introduce unrealistic boundary artifacts (Olsson et al., 2021; Tranheden et al., 2021). SA-ClassMix transfers each source-domain parcel together with its associated boundary neighborhood as a unified structural unit. This operation preserves the internal parcel-boundary relationship within the transferred source region, rather than independently mixing parcel interiors and boundary labels. As shown in Fig. 15, source parcel structures are embedded into target scenes while retaining their associated boundary context and coherent internal shapes. However, the mixing operation does not formally guarantee global boundary closure or continuity at the source-target interface.

The qualitative results show that SA-ClassMix promotes more complete parcel instances and sharper boundaries under domain shifts. Compared with appearance-level mixing, this structure-aware strategy better matches the mapping requirement of terraced landscapes, where parcel interiors and boundaries should be interpreted as coupled spatial units.

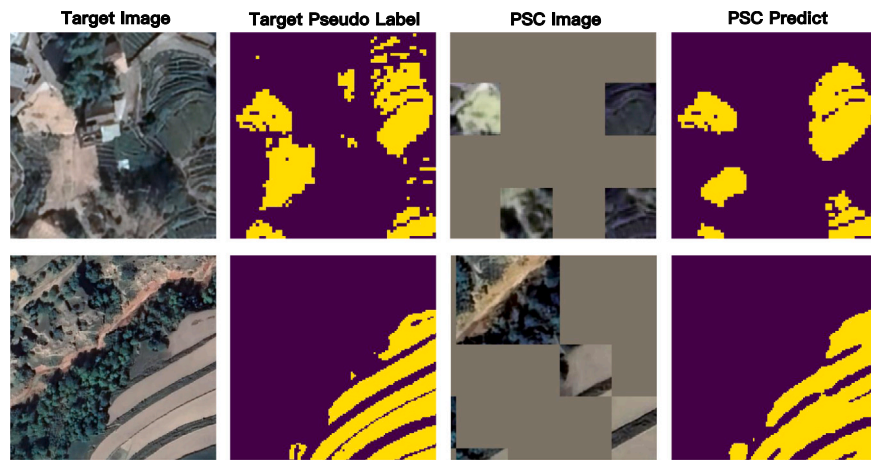


Fig. 14. Visualization of masked contextual reconstruction under PSC. Given partially masked target images, the student network recovers spatially coherent parcel regions from surrounding context, illustrating reduced reliance on local appearance cues.

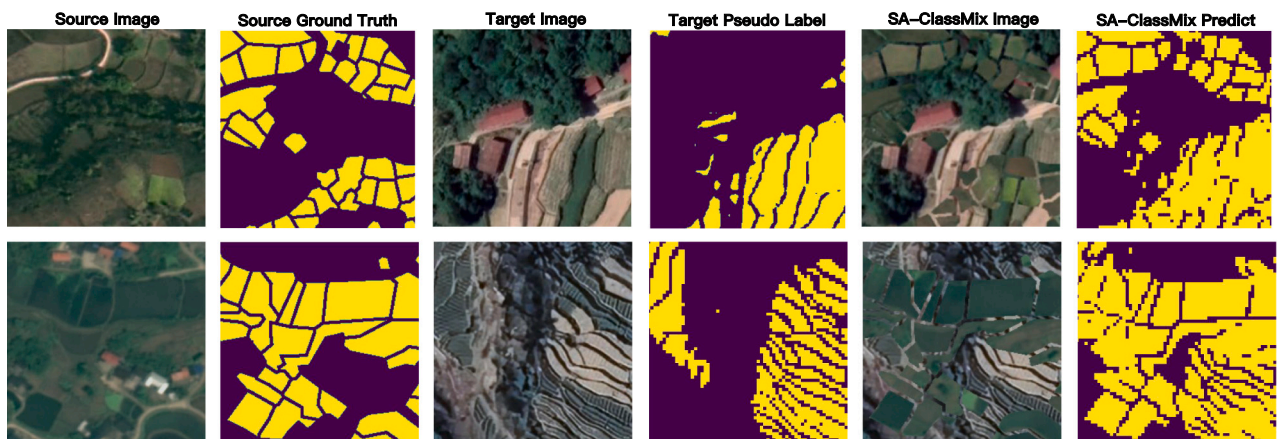


Fig. 15. Visualization of structure-preserving domain mixing with SA-ClassMix. Source-domain parcels and their associated boundary neighborhoods are embedded into target scenes as unified structural units, promoting internally coherent parcel predictions under domain shifts.

5.7. Extensibility and limitations for earth observation applications

DATerra is currently designed and evaluated using very-high-resolution RGB optical imagery. Additional observations may provide complementary information for terraced parcel extraction. In particular, DEM-derived elevation, slope, and aspect could help distinguish true terrace structures from visually similar ridges, roads, and shadows (Y. Li et al., 2025). SAR and multi-temporal optical imagery may provide additional structural and seasonal information (Hamidi et al., 2024), while NIR and hyperspectral data may improve the discrimination of spectrally confusing land-cover types (Hong et al., 2022; Roy et al., 2025). However, these modalities have not been evaluated in this study, and their integration would require careful spatial registration, resolution matching, and multimodal feature fusion.

The current quantitative experiments on GTPBD-China, the labeled cross-country scenes, and FHAPD are all based on RGB optical imagery. Therefore, the reported results support cross-regional and cross-dataset transferability within the RGB setting, but should not be interpreted as evidence of robustness across sensors or spectral modalities. The evaluated datasets do not provide co-registered DEM, SAR, NIR, hyperspectral, or multi-temporal observations with consistent parcel and boundary annotations. Quantitative evaluation of these data sources is therefore left for future work.

Although DATerra does not require parcel or boundary annotations for the target domain, its internally generated boundary cues may become less reliable when parcel boundaries are severely obscured by

Table 6

Scene-level cross-country validation on six unseen country scenes from GTPBD. All scenes follow the same imagery-quality criteria, preprocessing and tiling pipeline, spatial-resolution range, and annotation protocol as GTPBD-China. Values are percentages.

Country/Scene	F1↑	OA↑	mIoU↑	OIS↑	ODS↑	GUC↓
DPRK	78.68	73.44	60.37	79.01	62.53	13.22
Philippines	76.48	94.14	77.73	80.89	64.36	25.93
Myanmar	79.67	76.57	61.45	69.58	52.01	1.38
Nepal	72.41	86.56	70.22	88.74	73.30	7.49
Indonesia	84.89	84.65	73.38	93.01	82.65	9.53
Vietnam	82.41	83.13	71.09	87.08	73.33	5.16

vegetation, shadows, image degradation, or insufficient spatial resolution.

6. Scene-level cross-country validation and large-area inference on GTPBD

To quantitatively evaluate geographic transferability beyond the China-centered experiments, we select six labeled scenes from the global GTPBD dataset, covering the DPRK, the Philippines, Myanmar, Nepal, Indonesia, and Vietnam. These scenes follow the same imagery-quality criteria, preprocessing and tiling pipeline, spatial-resolution range, and parcel-boundary annotation protocol as GTPBD-China. None

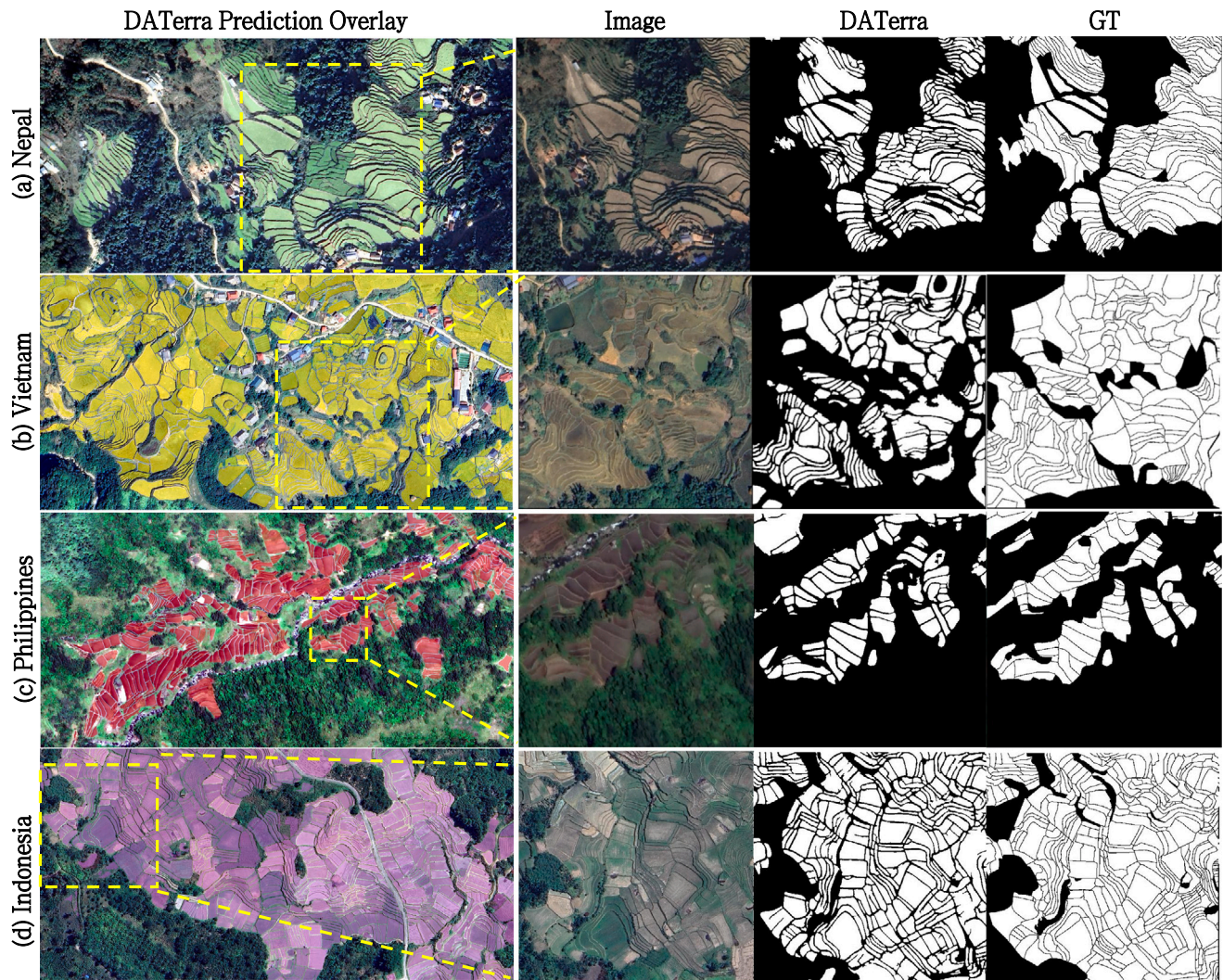


Fig. 16. Scene-level cross-country validation and large-area inference of DATerra on representative unseen GTPBD scenes from (a) Nepal, (b) Vietnam, (c) the Philippines, and (d) Indonesia. For each country, the columns from left to right show the DATerra prediction overlay, the VHR optical image, the DATerra prediction, and the GTPBD reference label. White denotes parcel interiors, whereas black denotes background or inter-parcel boundaries. None of these country scenes was used for training, validation, checkpoint selection, or country-specific fine-tuning.

of them was involved in model training, validation, checkpoint selection, or country-specific fine-tuning. Fig. 16 presents the input imagery, GTPBD reference labels, and DATerra predictions, while Table 6 reports the corresponding quantitative results. DATerra obtains F1 scores of 72.41%–84.89%, mIoU values of 60.37%–77.73%, and OIS values of 69.58%–93.01% across the six scenes. These results support transferability to the evaluated unseen geographic scenes under a consistent data and annotation protocol. Nevertheless, because each country is represented by only one selected scene, the results should be interpreted as scene-level cross-country validation rather than country-wide or global generalization.

Beyond demonstrating geographic transferability, these results also highlight the practical relevance of scalable terraced parcel mapping. Terraced agriculture is widely distributed across mountainous regions, where it supports food production, slope farming, and soil and water conservation (Ferro-Vázquez et al., 2017). In many of these regions, detailed parcel records remain incomplete and repeated field surveys are costly, making parcel mapping from remote sensing imagery particularly valuable for agricultural management (Waldner and Diakogiannis, 2020). Accurately delineated parcels with reliable boundaries and parcel-level integrity provide fundamental spatial units for parcel counting, area aggregation, parcel-level crop classification, yield

prediction, land-management inventories, and cross-temporal change monitoring (Weiss et al., 2020; Graesser and Ramankutty, 2017). Complete parcel boundaries also allow multispectral and multi-temporal observations to be associated with the correct agricultural management units and reduce spectral and temporal feature mixing between adjacent fields, thereby supporting more reliable parcel-level feature aggregation. Therefore, scalable cross-regional terraced parcel extraction can serve as a foundational step for transforming remote sensing observations into agricultural information in data-scarce mountainous regions. Further integrating DATerra into downstream parcel-based agricultural workflows represents an important direction for future research.

7. Conclusions

In this study, we presented DATerra, a boundary- and structure-aware domain adaptation framework for cross-regional terraced parcel extraction. The framework addresses two key challenges in terraced Earth Observation mapping: boundary degradation and structural inconsistency under domain shifts. By integrating geometry-aware feature enhancement, uncertainty-guided boundary refinement, parcel structural consistency learning, and structure-aware cross-domain mixing, DATerra improves parcel delineation while preserving the intrinsic

relationship between parcel interiors and enclosing boundaries. Extensive experiments on the GTPBD-China benchmark, covering twelve cross-regional transfer directions, demonstrate the effectiveness of the proposed framework. DATerra achieves clear gains in boundary- and object-level metrics, including improvements of 14.0% and 10.76% in OIS and ODS, together with reductions of 20.14% and 9.98% in GUC and GTC, respectively, compared with baseline UDA methods. The method also delivers consistent pixel-level improvements, with an average gain of 1.23% across transfer scenarios. These results indicate that explicitly enhancing boundary geometry and parcel-level structural consistency is important for robust terraced parcel extraction under severe regional heterogeneity. Beyond GTPBD-China, scene-level cross-country validation on six unseen GTPBD scenes further demonstrates the potential of DATerra for terraced parcel mapping outside the training regions. Without country-specific fine-tuning, DATerra maintains consistent pixel-, boundary-, and object-level performance across geographically diverse landscapes, highlighting its promising transferability for broader Earth Observation applications.

CRedit authorship contribution statement

Zhiwei Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Xingyuan Zeng:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Xinkai Kong:** Writing – review & editing, Writing – original draft. **Juepeng Zheng:** Writing – review & editing, Writing – original draft, Software, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Haoyuan Liang:** Writing – review & editing. **Guowei Zhou:** Writing – original draft. **Bohan Shi:** Resources. **Lingyuan Zhao:** Software, Data curation. **Jianxi Huang:** Resources, Funding acquisition, Data curation. **Haohuan Fu:** Supervision, Project administration.

Consent to participate

Not applicable.

Consent for publication

All authors have read and approved the final version of the manuscript and agree to its submission to International Journal of Applied Earth Observation and Geoinformation.

Ethics approval

Not applicable. This study does not involve human participants, human data, or animal experiments.

Use of artificial intelligence

During the preparation of this manuscript, AI-assisted tools were used only for language polishing and editing. The authors reviewed and edited the content as needed and take full responsibility for the content of the submitted manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The GTPBD dataset, including the GTPBD-China domains and the labeled cross-country scenes used in this study, is publicly available on Hugging Face at <https://huggingface.co/datasets/wxqzzw/GTD>. The Henan and Jiangsu subsets of FHAPD used for external-benchmark validation are publicly available through the official HBGNet repository at <https://github.com/NanNanmei/HBGNet>. The code is accessible at <https://github.com/Z-ZW-WXQ/DATerra>.

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